

Triage Learning : Human-AI Collaboration In Medical Diagnosis

A Thesis Submitted in Partial Fulfilment of the Requirements
for the Degree of

MASTER OF TECHNOLOGY in Artificial Intelligence & Data Science

by

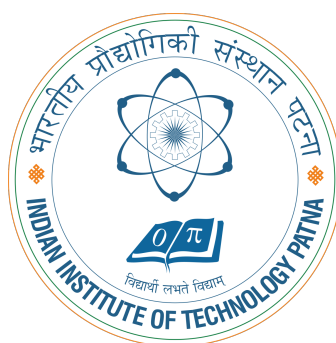
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Shanu Prakash

Indian Institute of Technology Patna

May 2026

CERTIFICATE

This is to certify that the M.Tech thesis titled “**Triage Learning : Human-AI Collaboration In Medical Diagnosis**”, submitted by **Shanu Prakash** (Roll No.: 24A03RES171), in partial fulfilment of the requirements for the degree of **Master of Technology in Artificial Intelligence & Data Science** at the **Indian Institute of Technology Patna**, is a record of original work carried out under supervision.

To the best of my knowledge, the work reported in this thesis is original and has not been submitted to any other university or institute for any degree or diploma.

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DECLARATION

I, **Shanu Prakash** (Roll No.: 24A03RES171), a student of M.Tech in Artificial Intelligence & Data Science at the Indian Institute of Technology Patna, hereby declare that the thesis titled “**Triage Learning : Human-AI Collaboration In Medical Diagnosis**” submitted by me for the partial fulfilment of the requirements for the award of the degree of Master of Technology is a record of original research work carried out by me under the supervision of **Gyanendra Kumar**.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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ABSTRACT

Medical diagnostic errors remain a persistent challenge in healthcare, with misdiagnoses contributing to significant patient morbidity and mortality. While deep learning models have demonstrated impressive performance in medical image classification, their deployment in clinical settings is hindered by a lack of uncertainty quantification and an inability to recognise when predictions may be unreliable. This thesis presents a comprehensive framework for *uncertainty-aware triage learning* that intelligently combines AI predictions on confident cases with human expert review of uncertain cases, optimising the accuracy-automation trade-off through principled threshold selection.

Four convolutional and transformer-based architectures (ResNet18, DenseNet121, EfficientNet-B3, and ViT-Small) are evaluated on three MedMNIST datasets: PathMNIST (histopathology, 9 classes), ChestMNIST (chest X-ray, 14 binary labels), and DermaMNIST (dermatoscopy, 7 classes). Monte Carlo Dropout and Deep Ensembles are compared as uncertainty quantification methods, and Temperature Scaling is applied for post-hoc calibration.

On PathMNIST, the baseline ResNet18 achieves 89.04% accuracy. Monte Carlo Dropout uncertainty attains a point-biserial correlation of 0.498 with prediction errors and an AUROC of 0.762 as an error detector. The threshold-based triage system improves accuracy to **97.35%**, reducing misdiagnoses from 787 to 190 — a **75.8% error reduction** — while automating 63.9% of cases. Temperature Scaling reduces Expected Calibration Error from 0.087 to 0.031. Deep Ensembles provide marginally higher triage accuracy (98.14%) at a fivefold computational cost. EfficientNet-B3 achieves the highest standalone accuracy (91.38%); its superiority over ResNet18, DenseNet121, and ViT-Small is confirmed by McNemar’s test after Bonferroni correction. Cross-dataset experiments confirm consistent triage benefit on ChestMNIST (+9.3 pp) and DermaMNIST (+10.4 pp). Grad-CAM analysis validates clinically plausible model attention and identifies systematic failure modes. Robustness experiments show that MC Dropout uncertainty increases monotonically under Gaussian noise, providing natural out-of-distribution detection.

The framework demonstrates that, for an institution processing 10,000 cases annually, the accuracy-optimal triage configuration prevents approximately 830 misdiagnoses per year while reducing cost from \$10.96 to \$4.53 per case, with a staffing requirement of 2–4 full-time equivalent reviewers. All results are supported by 95% bootstrap confidence intervals.

Keywords: Triage learning, uncertainty quantification, Monte Carlo Dropout, deep ensembles, medical image classification, human-AI collaboration, calibration, Grad-CAM, MedMNIST.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AR	Automation Rate
AUC	Area Under the Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BCC	Basal Cell Carcinoma
BCE	Binary Cross-Entropy
CAM	Class Activation Mapping
CNN	Convolutional Neural Network
DR	Deferral Rate
ECE	Expected Calibration Error
ELBO	Evidence Lower Bound
FTE	Full-Time Equivalent
GAP	Global Average Pooling
Grad-CAM	Gradient-weighted Class Activation Mapping
H&E	Haematoxylin and Eosin
HITL	Human-in-the-Loop
JSD	Jensen-Shannon Divergence
KL	Kullback-Leibler (divergence)
LR	Learning Rate
MBConv	Mobile Inverted Bottleneck Convolution
MC	Monte Carlo
MCE	Maximum Calibration Error
MedMNIST	Medical Modified National Institute of Standards and Technology
MHA	Multi-Head Attention
MSA	Multi-head Self-Attention
NLL	Negative Log-Likelihood
NLP	Natural Language Processing
RGB	Red, Green, Blue
ReLU	Rectified Linear Unit
ResNet	Residual Network

ROC	Receiver Operating Characteristic
SE	Squeeze-and-Excitation
SoTA	State of the Art
TCAV	Testing with Concept Activation Vectors
UQ	Uncertainty Quantification
ViT	Vision Transformer

LIST OF SYMBOLS

Symbol	Description
$\mathbf{x}$	Input image tensor
y	True class label
$\hat{y}$	Predicted class label
f_{θ}	Neural network with parameters θ
$\mathbf{z}$	Logit vector (pre-softmax activations)
$\hat{\mathbf{p}}$	Predicted class probability vector
$\bar{\mathbf{p}}$	Mean MC Dropout probability vector
C	Number of classes
N	Number of test samples
T	Number of MC Dropout forward passes
M	Number of ensemble members
T^*	Optimal temperature scaling parameter
τ	Uncertainty deferral threshold
τ^*	Optimal deferral threshold
d_i	Deferral indicator for sample i
u_i	Uncertainty score for sample i
$H[\cdot]$	Shannon entropy
p_h	Simulated human expert accuracy
w_c	Class weight for class c in weighted loss
n_c	Number of training samples in class c
r_{pb}	Point-biserial correlation coefficient
ρ_s	Spearman rank correlation coefficient
c_{err}	Cost per AI misclassification error
c_{rev}	Cost per human review instance
C_{total}	Total operational cost
α_k^c	Grad-CAM importance weight for feature map k , class c
A^k	k -th convolutional feature map
L^c	Grad-CAM localisation map for class c
μ	ImageNet normalisation mean [0.485, 0.456, 0.406]
σ	ImageNet normalisation std [0.229, 0.224, 0.225]

η	Learning rate
λ	Weight decay coefficient

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The integration of artificial intelligence into clinical medicine represents one of the most consequential technological transitions in modern healthcare. Deep learning models trained on large annotated datasets have demonstrated diagnostic accuracy comparable to, and in some domains exceeding, trained specialists across tasks ranging from diabetic retinopathy screening [5] to histopathological cancer detection [6]. Yet despite a decade of impressive benchmark results, clinical adoption of AI-driven diagnostic tools remains limited. A fundamental barrier is not one of accuracy, but of trust: standalone AI systems cannot communicate when they are uncertain, cannot acknowledge the limits of their own competence, and cannot integrate gracefully with human clinical workflows.

Diagnostic error in medicine is a persistent and costly problem. Studies estimate that 5–15% of patient encounters involve some form of diagnostic inaccuracy, contributing to an estimated 40,000–80,000 deaths annually in the United States alone [7]. In pathology and radiology, where visual interpretation of complex images is central, error rates of 2–5% under routine workload conditions have been documented even among experienced specialists [8]. Cognitive fatigue, high case volumes, and rare class presentations compound these risks. Automated systems that handle routine, clearly-defined cases accurately—while escalating ambiguous, edge-case samples to human review—present a compelling alternative to both fully manual and fully automated paradigms.

The concept of *triage learning*, or learning to defer, formalises this intuition. Rather than requiring a model to produce a final prediction for every input, a triage system partitions cases into those the model handles confidently and those referred to a human expert, with the allocation governed by a principled measure of predictive uncertainty. The combined human-AI system can achieve accuracy well above either component operating independently, provided the uncertainty signal is reliable and the deferral threshold is optimised appropriately.

Uncertainty quantification (UQ) is therefore a prerequisite for effective triage. A model that cannot distinguish its own confident correct predictions from its confident errors provides no useful signal for routing decisions. Monte Carlo

(MC) Dropout [9], Deep Ensembles [10], and post-hoc calibration methods such as temperature scaling [11] collectively provide a toolkit for eliciting and refining probabilistic uncertainty estimates from neural networks. These methods differ in computational cost, theoretical grounding, and empirical reliability, motivating systematic comparison within a unified experimental framework.

Model interpretability presents a complementary concern. Even when an AI system achieves high accuracy, clinicians need explanations of which visual features drive predictions in order to identify systematic failure modes, build appropriate trust, and satisfy regulatory requirements. Gradient-weighted Class Activation Mapping (Grad-CAM) [12] provides spatially-resolved saliency maps that localise discriminative regions within input images, enabling qualitative assessment of whether model reasoning aligns with known pathological features.

This thesis addresses these interrelated challenges—uncertainty quantification, triage policy design, calibration, interpretability, and cross-dataset generalisation—within a single cohesive experimental framework applied to medical image classification. The work proceeds from a validated baseline on histopathology tissue classification to a comprehensive multi-dataset, multi-architecture, multi-UQ-method evaluation that systematically characterises where and why human-AI collaboration improves upon standalone AI.

1.2 Problem Statement

Consider a clinical AI system deployed to classify medical images into diagnostic categories. The system processes a stream of incoming cases and must decide, for each case, whether to (a) issue an automated prediction with high confidence, or (b) defer the case to a human clinician for expert review. The central challenge admits three inter-related sub-problems:

1. **Uncertainty estimation:** Given a trained classifier f_θ and input image x , compute a scalar uncertainty score $u(x) \in \mathbb{R}_{\geq 0}$ such that cases where f_θ is likely to err receive high $u(x)$ and cases where f_θ is reliable receive low $u(x)$.
2. **Threshold optimisation:** Given an uncertainty-to-error correlation, identify the deferral threshold τ^* that maximises a combined objective incorporating system accuracy, automation rate, and operational cost.
3. **Calibration and reliability:** Ensure that predicted class probabilities are well-calibrated—i.e., that a predicted probability of 0.9 corresponds to approximately 90% empirical accuracy—so that uncertainty scores derived from model outputs are meaningful rather than artefactual.

These three sub-problems interact: a poorly calibrated model may produce unreliable uncertainty scores, leading to a suboptimal triage boundary even after threshold sweep. Improving calibration through temperature scaling alters the uncertainty distribution and shifts the optimal threshold. Switching from MC Dropout to Deep Ensembles changes both the accuracy of the base model and the quality of uncertainty estimation, requiring joint evaluation.

The evaluation context introduces further constraints: medical datasets exhibit severe class imbalance, images may be low-resolution (28×28 pixels in the MedMNIST benchmark), task types vary from single-label multi-class classification to multi-label binary classification, and generalisation across imaging modalities (histopathology, chest radiography, dermatoscopy) cannot be assumed.

1.3 Objectives

The specific objectives of this research are as follows:

1. Train and evaluate baseline deep learning classifiers—ResNet18, DenseNet121, EfficientNet-B3, and Vision Transformer (ViT-Small)—on the PathMNIST histopathology dataset and compare their classification accuracy, calibration error, and training efficiency.
2. Implement Monte Carlo Dropout uncertainty estimation and quantify the correlation between model uncertainty and prediction errors, validating uncertainty as a viable signal for triage decisions.
3. Design and evaluate a threshold-based deferral triage system on PathMNIST, identifying optimal thresholds under accuracy-maximising and cost-minimising objectives, and characterising system performance under varying human reviewer accuracy assumptions.
4. Improve probability calibration using temperature scaling and assess its impact on Expected Calibration Error (ECE), reliability diagrams, and triage system performance.
5. Compare MC Dropout against Deep Ensembles for uncertainty quality, examining whether ensemble diversity provides superior uncertainty-error correlation and triage benefit.
6. Extend the triage framework to ChestMNIST (chest X-ray multi-label classification) and DermaMNIST (dermatoscopy multi-class classification) to assess cross-dataset and cross-modality generalisation.
7. Generate Grad-CAM visualisations for representative correctly classified and

misclassified samples to interpret model attention and identify failure modes.

8. Conduct robustness analysis under additive Gaussian noise and mild distribution shift to characterise how uncertainty scores respond to out-of-distribution inputs.
9. Perform statistical validation of performance comparisons using bootstrap confidence intervals and McNemar’s test for significance of pairwise accuracy differences.

1.4 Scope and Contributions

1.4.1 Scope

This work is scoped to image classification tasks within the MedMNIST [1] benchmark collection, which provides standardised preprocessing, splits, and evaluation protocols across eleven two-dimensional imaging datasets. Three datasets are studied: PathMNIST (histopathology), ChestMNIST (chest radiography), and DermaMNIST (dermatoscopy). All experiments use the standard 28×28 pixel resolution of MedMNIST; experiments at 224×224 resolution are not included due to computational constraints. Human experts are simulated probabilistically rather than through a prospective user study with clinical participants. While simulation allows systematic sensitivity analysis across a continuous range of human accuracy levels, it does not capture the full complexity of human diagnostic reasoning or human-AI interaction dynamics.

1.4.2 Contributions

The principal contributions of this thesis are:

1. **End-to-end triage learning framework.** A complete and reproducible methodology spanning model training, Monte Carlo Dropout and Deep Ensemble uncertainty estimation, temperature scaling calibration, threshold optimisation under multiple objectives, and cost-benefit analysis. The framework is modular and directly applicable to new datasets and architectures.
2. **Empirical validation of uncertainty-based triage on histopathology.** Demonstrated that MC Dropout uncertainty achieves a point-biserial correlation of 0.498 with prediction errors on PathMNIST, enabling a triage system that improves accuracy from 89.04% to 97.35%—a 75.8% reduction in misdiagnoses—while automating 63.9% of cases.
3. **Comparative evaluation of uncertainty quantification methods.** Systematic comparison of MC Dropout and Deep Ensembles across accuracy,

calibration, uncertainty quality, and computational overhead, providing empirical guidance for UQ method selection in resource-constrained clinical deployments.

4. **Calibration analysis and temperature scaling study.** Characterisation of baseline model calibration quality via ECE and reliability diagrams, with post-hoc temperature scaling shown to reduce ECE substantially, and analysis of the downstream impact on triage threshold and system accuracy.
5. **Multi-dataset generalisation study.** Extension of the triage framework to chest radiography and dermatoscopy datasets, providing evidence that the methodology generalises across imaging modalities, class structures (multi-label vs. multi-class), and disease prevalence patterns.
6. **Interpretability analysis via Grad-CAM.** Qualitative assessment of model attention on PathMNIST and DermaMNIST, relating discriminative regions to known histological and dermatological features and identifying cases where saliency diverges from clinical expectation as indicators of systematic error modes.
7. **Robustness and statistical validation.** Quantification of triage benefit under additive noise and mild covariate shift, together with bootstrap confidence intervals and significance tests for all reported accuracy comparisons, ensuring that conclusions are statistically defensible.

1.5 Organisation of the Thesis

The remainder of this thesis is organised as follows.

Chapter 2 provides a structured review of the relevant literature, covering deep learning architectures for medical image analysis, uncertainty quantification methods, human-AI collaboration frameworks, model interpretability, and probability calibration. The chapter concludes with a synthesis of research gaps that motivate the present work.

Chapter 3 describes the three datasets employed in the study—PathMNIST, ChestMNIST, and DermaMNIST—detailing their provenance, class distributions, known challenges, and the preprocessing and augmentation pipelines applied. Evaluation metrics and the experimental protocol are also defined.

Chapter 4 presents the methodology in full, covering all model architectures, the MC Dropout and Deep Ensemble UQ frameworks, temperature scaling calibration, the triage deferral policy, threshold optimisation procedures, the cost modelling framework, and the Grad-CAM interpretability pipeline.

Chapter 5 presents all experimental results, organised from baseline model performance through uncertainty analysis, triage optimisation, calibration study, cross-dataset generalisation, interpretability visualisations, and robustness experiments, concluding with statistical validation.

Chapter 6 interprets and discusses the results in the context of related work, addresses limitations and failure modes, and considers implications for clinical deployment.

Chapter 7 summarises the contributions of the thesis and outlines directions for future research.

Appendices provide supplementary materials including complete hyperparameter configurations, extended per-class performance tables, and notes on code structure and reproducibility.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Deep Learning in Medical Image Analysis

2.1.1 Convolutional Neural Network Architectures

The modern era of deep learning in medical imaging began with the application of Convolutional Neural Networks (CNNs) to large-scale image recognition. The seminal AlexNet [13] demonstrated that deep convolutional architectures trained on ImageNet could surpass classical feature engineering by a substantial margin. Subsequent architectural innovations aimed at training deeper and more parameter-efficient networks, addressing the twin problems of vanishing gradients and representational bottlenecks.

He et al. [14] introduced Residual Networks (ResNets), which resolve the degradation problem in very deep networks by adding shortcut connections that allow gradients to flow directly across layers. A residual block computes $\mathcal{F}(\mathbf{x}) + \mathbf{x}$ where $\mathcal{F}$ represents the residual mapping to be learned. ResNet18 and ResNet50 became standard baselines across medical imaging benchmarks due to their favourable accuracy-to-parameter ratio and well-studied transfer learning behaviour. Huang et al. [15] extended this idea with Dense Convolutional Networks (DenseNets), where each layer receives concatenated feature maps from all preceding layers, enabling feature reuse and reducing the total number of parameters required. DenseNet121 has achieved strong results on chest radiograph classification and histopathology tasks.

Tan and Le [16] proposed a principled compound scaling methodology that jointly scales network width, depth, and input resolution using a fixed set of coefficients. The resulting EfficientNet family achieves state-of-the-art accuracy with substantially fewer parameters than comparable architectures. EfficientNet-B3 and B4 are particularly well-suited to medical imaging tasks where computational budgets are constrained and datasets are moderate in size.

Dosovitskiy et al. [17] introduced the Vision Transformer (ViT), which abandons convolutions entirely in favour of patch-based self-attention. An image is divided into fixed-size patches, linearly projected into embedding vectors, and processed by a standard Transformer encoder with multi-head self-attention. ViT achieves competitive or superior performance to CNNs when trained on sufficiently large

datasets. In the medical imaging domain, where labelled data is often limited, ViT variants pre-trained with self-supervised objectives [18] have shown promise, though their advantage over CNNs at low resolution (28×28 pixels) remains less established.

2.1.2 Transfer Learning and Domain Adaptation

Transfer learning from large-scale natural image datasets to medical imaging is now standard practice. Raghu et al. [19] conducted a systematic study of transfer learning for medical imaging and found that features from ImageNet pre-training transfer effectively to histopathology and radiology tasks, particularly in lower layers that capture texture and edge statistics. However, they noted that for sufficiently large medical datasets, training from scratch can approach or match transfer learning performance, suggesting the benefit is primarily one of sample efficiency rather than qualitative improvement.

Layer-wise learning rate decay—applying smaller learning rates to earlier layers and larger rates to task-specific classifier heads—has become standard when fine-tuning pre-trained models, preventing destructive forgetting of general features while allowing efficient adaptation to the target domain [20]. Gradual unfreezing, which progressively unfreezes layers from the classifier head backwards, provides a complementary strategy with similar motivation.

2.1.3 Medical Imaging Benchmarks and Datasets

Yang et al. [1] introduced the MedMNIST benchmark collection, comprising standardised 28×28 pixel versions of eleven two-dimensional and six three-dimensional medical imaging datasets spanning pathology, radiology, dermatology, ophthalmology, and dental imaging. The benchmark provides fixed train-validation-test splits, evaluation protocols, and baseline results, enabling direct comparison across studies. PathMNIST contains 89,996 histopathology images across nine colorectal cancer tissue classes sourced from Kather et al. [2]. ChestMNIST is derived from the NIH ChestX-ray14 dataset [3] with 112,120 frontal-view chest radiographs and 14 binary disease labels. DermaMNIST is sourced from the HAM10000 skin lesion dataset [4] with 10,015 dermatoscopic images across seven diagnostic categories.

2.1.4 Class Imbalance in Medical Datasets

Severe class imbalance is a pervasive challenge in medical imaging datasets, arising naturally from the rarity of pathological conditions relative to normal tissue. Buda et al. [21] conducted a systematic study of imbalance mitigation strategies and found

that oversampling the minority class at the data level and using cost-sensitive weighting at the loss level both provide substantial benefits, with the optimal strategy dependent on the degree of imbalance. Weighted cross-entropy assigns per-class weight $w_c = n/(C \cdot n_c)$, inversely proportional to class frequency. Lin et al. [22] proposed focal loss, which dynamically down-weights easy, well-classified examples by multiplying cross-entropy by $(1 - p_t)^\gamma$, focusing training on hard minority-class examples.

2.1.5 Histopathology and Dermatology Classification

Esteva et al. [6] demonstrated that a deep CNN trained on 129,450 clinical images could classify skin cancer at a level of competence comparable to board-certified dermatologists. Campanella et al. [23] showed that weakly supervised learning on over a million whole-slide images could achieve clinically actionable performance for prostate cancer detection. In histopathology, Kather et al. [2] demonstrated that deep learning classifiers could predict microsatellite instability directly from haematoxylin-and-eosin stained tissue images, illustrating the potential for discovery of novel biomarkers beyond the reach of manual pathological assessment.

2.2 Uncertainty Quantification in Neural Networks

2.2.1 Bayesian Deep Learning

Standard neural networks produce point estimates of parameters and, consequently, point estimates of predictions. From a Bayesian perspective, one seeks instead the posterior predictive distribution $p(y | \mathbf{x}, \mathcal{D}) = \int p(y | \mathbf{x}, \boldsymbol{\theta}) p(\boldsymbol{\theta} | \mathcal{D}) d\boldsymbol{\theta}$, which marginalises over parameter uncertainty given the training data $\mathcal{D}$. Exact Bayesian inference is intractable for modern deep networks due to the high dimensionality of $\boldsymbol{\theta}$. Variational inference approximates the posterior with a tractable distribution $q_\phi(\boldsymbol{\theta})$ by minimising the KL divergence $\text{KL}[q_\phi(\boldsymbol{\theta}) || p(\boldsymbol{\theta} | \mathcal{D})]$, equivalent to maximising the evidence lower bound (ELBO). Blundell et al. [24] introduced Bayes by Backprop, which learns factorised Gaussian posteriors over weights. While theoretically principled, weight-space variational methods incur a twofold memory and computational overhead.

2.2.2 Monte Carlo Dropout

Gal and Ghahramani [9] showed that training a neural network with dropout and evaluating it at test time with dropout active is equivalent to performing approximate variational inference in a deep Gaussian process, provided the variational distribution is a mixture of two Gaussians with zero mean. This reinterpretation

tion, termed Monte Carlo (MC) Dropout, provides a computationally lightweight uncertainty estimate: T stochastic forward passes through the network with active dropout yield a set of predictions $\{\hat{\mathbf{p}}_t\}_{t=1}^T$, from which the mean prediction $\bar{\mathbf{p}} = T^{-1} \sum_t \hat{\mathbf{p}}_t$ and predictive entropy $H[\bar{\mathbf{p}}] = -\sum_c \bar{p}_c \log \bar{p}_c$ are computed.

MC Dropout has been applied extensively in medical imaging. Leibig et al. [25] found that MC Dropout uncertainty outperformed conventional confidence measures for diabetic retinopathy detection, achieving AUROCs of 0.82–0.89 for uncertainty-based error prediction. Roy et al. [26] demonstrated that MC Dropout uncertainty maps in cardiac MRI segmentation closely correspond to anatomically ambiguous boundaries, providing actionable quality control information. The key limitation of MC Dropout is that the Bayesian interpretation requires dropout rates and architecture choices made originally for regularisation, not posterior approximation, which may produce poorly calibrated uncertainties.

2.2.3 Deep Ensembles

Lakshminarayanan et al. [10] proposed Deep Ensembles as a simple, scalable, and highly effective approach to uncertainty estimation. An ensemble of M independently initialised and trained models produces predictions $\{\hat{\mathbf{p}}^{(m)}\}_{m=1}^M$; uncertainty is captured by the disagreement among ensemble members, typically measured by the entropy of the mean prediction or the average pairwise Jensen-Shannon divergence. Deep ensembles consistently outperform MC Dropout on calibration benchmarks and tend to produce superior uncertainty-error correlations, particularly on out-of-distribution inputs [27]. Ovadia et al. [27] showed that ensemble uncertainty is more robust to distribution shift than MC Dropout uncertainty, an important property for medical applications where test distributions may diverge from training distributions due to scanner variability or population differences.

The principal drawback of deep ensembles is computational cost: training M models requires M times the computational budget, and inference requires M forward passes. For $M = 5$, the standard ensemble size in the literature, this is typically acceptable for batch processing applications but may constrain deployment in latency-sensitive real-time systems.

2.2.4 Other Uncertainty Estimation Methods

Evidential deep learning [28] places a Dirichlet distribution over the class probability simplex, treating the network as estimating the parameters of an evidential distribution rather than directly outputting class probabilities. This allows direct estimation of epistemic uncertainty (lack of knowledge) and aleatoric un-

certainty (irreducible noise) without ensemble or dropout. Spectral normalisation with Gaussian processes [29] regularises neural network features to respect a bi-Lipschitz constraint, ensuring that the output feature space is well-behaved for downstream distance-based uncertainty estimation. Conformal prediction [30] provides distribution-free coverage guarantees: given a miscoverage level α , the prediction set constructed from a conformal score is guaranteed to contain the true label with probability at least $1 - \alpha$, regardless of the underlying data distribution. These methods collectively define the landscape within which MC Dropout and Deep Ensembles should be understood.

2.3 Human-AI Collaborative Systems and Learning to Defer

2.3.1 Selective Prediction and Rejection

The classical selective prediction framework, introduced by Chow [31], allows a classifier to abstain on inputs where its confidence falls below a threshold, trading coverage for selective accuracy. Geifman and El-Yaniv [32] extended this to neural networks, proposing selective classifiers based on softmax response thresholds and demonstrating empirically that competitive accuracy-coverage trade-offs are achievable without modifying training. Selective prediction provides the theoretical foundation for triage systems: the AI system operates as a selective classifier, and deferred cases are handled by a human expert rather than simply rejected.

2.3.2 Learning to Defer

Madras et al. [33] introduced the predict-then-defer framework, in which a model is jointly trained to predict and to defer, with deferral cost depending on the human expert’s predicted accuracy for each input. Mozannar and Sontag [34] provided consistent surrogate loss functions for joint optimisation of AI accuracy on automated cases and human accuracy on deferred cases, showing that surrogate-minimising classifiers are consistent with the desired Bayes-optimal deferral policy. Verma and Nalisnick [35] showed that calibrated uncertainty scores enable consistent learning to defer without joint training, allowing post-hoc construction of deferral policies from existing trained classifiers. This result is directly relevant to the present work, which employs post-hoc threshold optimisation rather than end-to-end joint training.

Okati et al. [36] proposed differentiable triage, which directly minimises a smooth approximation of the system-level error using gradient-based optimisation, enabling more flexible deferral policies that can incorporate budget constraints and variable human expertise. The limitation of joint learning approaches, compared to post-

hoc threshold optimisation, is their requirement for human label estimates during training, which are rarely available in standard medical imaging datasets.

2.3.3 Human-AI Collaboration in Clinical Settings

Sendak et al. [37] conducted a qualitative study of ML deployment for mortality prediction in a real emergency department, finding that effective human-AI collaboration required workflow integration, explanation of model behaviour, and active trust calibration. Clinicians who trusted the model blindly performed worse than those who maintained appropriate scepticism. Reyes et al. [38] surveyed the interpretability requirements for clinical AI across six specialties, noting that clinicians' primary concern was not prediction accuracy but understanding why a prediction was made and under what conditions it could be trusted.

Tschandl et al. [39] demonstrated in a prospective study of dermatology diagnosis that AI assistance improved the diagnostic accuracy of general practitioners by 10 percentage points but did not improve the accuracy of experienced dermatologists, suggesting that triage benefit depends strongly on the relative competency gap between AI and human. Rajpurkar et al. [40] reviewed the state of AI in medical imaging and concluded that human-AI collaboration, rather than replacement, represented the near-term clinical reality, with AI systems best suited to augmenting human decision-making in high-volume routine tasks. Fernandes et al. [41] reviewed triage automation in emergency settings, reporting automation rates of 40–80% depending on acceptable accuracy targets, consistent with the operating range studied in this thesis.

2.3.4 Automation Bias and Trust Calibration

A well-documented failure mode of human-AI collaboration is automation bias: the tendency for humans to over-rely on AI recommendations, reducing critical scrutiny even when the AI is incorrect [42]. Automation bias is stronger when the human is under time pressure, when the AI recommendation is presented prominently, and when the human has limited domain expertise. In medical imaging, automation bias has been documented in radiological interpretation studies where AI annotations were shown alongside images. Uncertainty communication—explicitly displaying model confidence or flagging low-confidence cases—has been shown to partially counteract automation bias by prompting increased scrutiny on uncertain predictions [43].

2.4 Model Interpretability and Explainability

2.4.1 Gradient-Based Saliency Methods

Interpretability methods for neural networks can be classified along two axes: global versus local, and gradient-based versus perturbation-based. Local gradient-based methods explain individual predictions by computing the gradient of the output score with respect to the input image, highlighting pixels that most influence the prediction. Simonyan et al. [44] proposed the vanilla gradient approach, which computes $\partial S_c / \partial \mathbf{x}$ where S_c is the class score. Guided backpropagation [45] modifies the backpropagation pass to zero out negative gradients at ReLU activations, producing sharper and more spatially coherent saliency maps.

Integrated Gradients [46] addresses the saturation problem of vanilla gradients by accumulating gradients along a straight-line path from a baseline input $\mathbf{x}'$ to the actual input $\mathbf{x}$: $\text{IG}_i(\mathbf{x}) = (x_i - x'_i) \int_0^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} d\alpha$. This approach satisfies desirable axioms including sensitivity and implementation invariance, making it more theoretically principled than vanilla gradient methods.

2.4.2 Class Activation Mapping

Class Activation Mapping (CAM) [47] computes a spatial map indicating which image regions are most relevant to the predicted class by projecting classification weights back onto the final convolutional feature maps. For a network with global average pooling, the class activation map for class c is $M_c(\mathbf{x}) = \sum_k w_k^c A_k(\mathbf{x})$, where A_k is the k -th feature map and w_k^c is the weight connecting feature k to class c . CAM is limited to architectures with specific global average pooling structures.

Selvaraju et al. [12] generalised this to arbitrary architectures with Gradient-weighted CAM (Grad-CAM), which uses the gradient of the class score with respect to the feature maps of the target convolutional layer as an importance weighting: $\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial S_c}{\partial A_{ij}^k}$. The resulting localisation map $L^c = \text{ReLU}(\sum_k \alpha_k^c A_k)$ is upsampled to input resolution and overlaid on the original image. Grad-CAM has been widely adopted in medical imaging as a tool for validating that models attend to clinically relevant regions. Grad-CAM++ [48] extends Grad-CAM by using a weighted combination of the partial derivatives, providing improved localisation of multiple occurrences of the same class within a single image.

2.4.3 Applications in Medical Image Interpretability

Rajpurkar et al. [49] demonstrated Grad-CAM localisation for pneumonia detection on chest radiographs, showing that the network correctly focused on consolidation

regions. Esteva et al. [6] used saliency maps to show that the skin cancer classifier attended to lesion borders and texture features consistent with dermatological diagnostic criteria. However, Rudin [50] has argued that post-hoc saliency explanations for black-box models can be misleading, as saliency maps reflect the gradient landscape of the learned function rather than the causal reasoning process. Adebayo et al. [51] proposed sanity checks showing that many saliency methods produce similar maps even for randomly initialised or label-randomised networks, raising concerns about whether they measure model behaviour or merely edge-detection in the input image. These critiques motivate the use of Grad-CAM primarily as a qualitative diagnostic tool rather than a definitive explanation.

2.4.4 Concept-Based Explanations

Beyond pixel-level saliency, concept-based explanations aim to provide higher-level, human-interpretable reasons for predictions. Testing with Concept Activation Vectors (TCAV) [52] quantifies the degree to which user-defined visual concepts (e.g., “nuclear enlargement”, “irregular border”) influence network predictions, by training linear classifiers to detect concept presence in the activation space. In histopathology, concept-based explanations have been shown to align with pathologists’ verbal reasoning more closely than pixel saliency maps [53].

2.5 Calibration of Probabilistic Classifiers

2.5.1 Calibration and Overconfidence

A probabilistic classifier is said to be well-calibrated if the predicted probability of class c matches the empirical frequency of class c occurring across inputs assigned that probability. Formally, $P(\hat{y} = y \mid \hat{p} = p) = p$ for all $p \in [0, 1]$. Guo et al. [11] demonstrated that modern neural networks are systematically overconfident: despite higher accuracy than their predecessors, post-ReLU deep networks trained with batch normalisation and weight decay exhibit predicted probabilities concentrated near 0 and 1 with empirical accuracy substantially below the predicted probability. This overconfidence is problematic for clinical deployment, where predicted probabilities inform treatment decisions, and for uncertainty-based triage, where inflated confidence reduces the discriminative power of uncertainty scores.

2.5.2 Calibration Metrics

Expected Calibration Error (ECE) partitions predictions into B equally-spaced confidence bins and computes the weighted average of the absolute difference

between mean confidence and accuracy within each bin:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (2.1)$$

where B_b is the set of predictions falling in bin b , $\text{acc}(B_b)$ is the fraction correctly classified, and $\text{conf}(B_b)$ is the mean predicted confidence. Maximum Calibration Error (MCE) replaces the weighted average with a maximum over bins, quantifying worst-case miscalibration. Naeini et al. [54] introduced the reliability diagram, which plots $\text{acc}(B_b)$ against $\text{conf}(B_b)$ for each bin; a perfectly calibrated model produces a diagonal line. The Brier score [55] provides a proper scoring rule that simultaneously measures both calibration and sharpness: $\text{BS} = n^{-1} \sum_i \sum_c (\hat{p}_{ic} - y_{ic})^2$.

2.5.3 Post-Hoc Calibration Methods

Platt scaling [56], originally proposed for support vector machine outputs, applies a sigmoid transformation to logits: $\hat{p} = \sigma(a \cdot f(\mathbf{x}) + b)$, where a and b are learned on a held-out calibration set by minimising negative log-likelihood. Temperature scaling [11] is a single-parameter special case: $\hat{p}_c = \text{softmax}(z_c/T)$, where $T > 1$ “softens” the distribution and reduces overconfidence. Temperature scaling leaves class predictions unchanged and does not affect accuracy; it only modifies confidence magnitudes. Guo et al. found temperature scaling to outperform more complex methods including histogram binning, isotonic regression, and Platt scaling on modern neural networks, attributing this to the single-parameter model’s lower variance and better generalisation from limited calibration data.

Dirichlet calibration [57] generalises temperature scaling to multi-class settings by fitting a full Dirichlet distribution over the probability simplex, allowing class-dependent calibration corrections. Ensemble-based calibration [10] exploits the observation that ensembles of separately trained models are inherently better calibrated than individual models, as the diversity of ensemble members prevents systematic overconfidence on any subset of inputs.

2.5.4 Calibration in Medical AI

Nixon et al. [58] proposed measuring calibration with adaptive binning schemes that maintain equal sample sizes per bin, addressing the sensitivity of fixed-width bins to confidence concentration near 1. In the medical imaging domain, poorly calibrated models have been shown to induce suboptimal treatment decisions when used in clinical decision support systems [59]. Calibration degradation under distribution shift—where test inputs differ systematically from training inputs—is

a particular concern in medical imaging due to scanner variability, site-specific acquisition protocols, and patient population heterogeneity [27].

2.6 Summary and Research Gaps

The reviewed literature collectively establishes several conclusions. First, transfer learning from ImageNet to medical imaging is effective and standard, with ResNet, DenseNet, and EfficientNet variants providing competitive baselines. Vision Transformers offer complementary inductive biases but require more data or self-supervised pre-training to match CNNs in low-resource settings. Second, MC Dropout is the most widely deployed uncertainty quantification method due to its simplicity and minimal computational overhead, but Deep Ensembles consistently provide superior calibration and out-of-distribution uncertainty. Third, triage systems that combine AI predictions with human review can substantially outperform either component alone, and post-hoc threshold optimisation using calibrated uncertainty scores provides a practically viable alternative to joint learning to defer approaches. Fourth, Grad-CAM provides useful qualitative interpretability for convolutional networks, though its reliability as a causal explanation requires validation against clinical ground truth. Fifth, temperature scaling is a simple and effective post-hoc calibration method that systematically reduces overconfidence without affecting accuracy.

The following research gaps motivate the present work:

1. **End-to-end system integration.** Existing studies typically evaluate one component in isolation—uncertainty estimation, triage policy, or calibration—without assessing interactions between components. For example, the impact of temperature scaling on triage threshold selection and system accuracy has not been systematically studied.
2. **Comparative UQ evaluation within a common framework.** Direct comparisons of MC Dropout and Deep Ensembles on the same dataset, with the same base architecture and identical triage framework, are rare. Most comparative studies vary multiple factors simultaneously, making it difficult to attribute performance differences.
3. **Multi-dataset generalisation with consistent methodology.** Studies that apply triage frameworks to multiple medical imaging modalities using identical experimental protocols are uncommon, limiting understanding of whether triage benefits are dataset-specific or broadly applicable.
4. **Statistical rigour in performance comparisons.** Many studies report

point estimates without confidence intervals or significance tests, making it difficult to assess whether reported improvements over baselines are reproducible or attributable to random variation.

5. **Robustness of uncertainty under distribution shift.** The behaviour of MC Dropout and ensemble uncertainty under additive noise and mild covariate shift in the medical imaging context has been studied primarily with natural image benchmarks, with limited medical domain evaluation.

This thesis addresses all five gaps through a comprehensive, reproducible experimental framework applied to three MedMNIST datasets, with controlled comparisons of UQ methods, systematic calibration analysis, and statistical validation of all reported results.

CHAPTER 3

DATASETS AND EXPERIMENTAL SETUP

3.1 MedMNIST Benchmark Overview

The MedMNIST benchmark collection [1] provides a standardised suite of biomedical image classification datasets in a unified, lightweight format. Each dataset is pre-processed to a resolution of 28×28 pixels per image, supplied with fixed train-validation-test splits, and accompanied by canonical evaluation metrics, enabling direct reproducibility and cross-study comparison without dataset-specific preprocessing decisions. The benchmark spans multiple imaging modalities, anatomical regions, and clinical tasks, making it well-suited for evaluating the generalisability of machine learning frameworks across diverse medical imaging conditions.

Three datasets from the MedMNIST v2 collection are employed in this thesis: PathMNIST for histopathological tissue classification, ChestMNIST for multi-label chest radiograph disease detection, and DermaMNIST for dermatoscopic skin lesion classification. These three were selected to ensure diversity along three key axes: imaging modality (optical microscopy, X-ray, and dermoscopy), classification task structure (single-label multi-class versus multi-label binary), and dataset scale (small, medium, and large). Table 3.1 summarises the key characteristics of each dataset; Figure 3.1 visualises the class distribution within each.

Table 3.1: Summary of MedMNIST Datasets Used in This Work

Dataset	Modality	Total N	Train	Val	Test	Classes
PathMNIST	Histopathology (RGB)	89,996	62,997	13,499	13,500	9 (single-label)
ChestMNIST	Chest X-ray (greyscale)	112,120	78,484	16,818	16,818	14 (multi-label)
DermaMNIST	Dermatoscopy (RGB)	10,015	7,010	1,503	1,502	7 (single-label)

All images are loaded as uint8 arrays in the range $[0, 255]$ and subsequently normalised during preprocessing. Greyscale ChestMNIST images are replicated across three channels to enable use of RGB-pretrained architectures without architectural modification. The standard MedMNIST splits are used throughout without modification, ensuring comparability with published baselines.

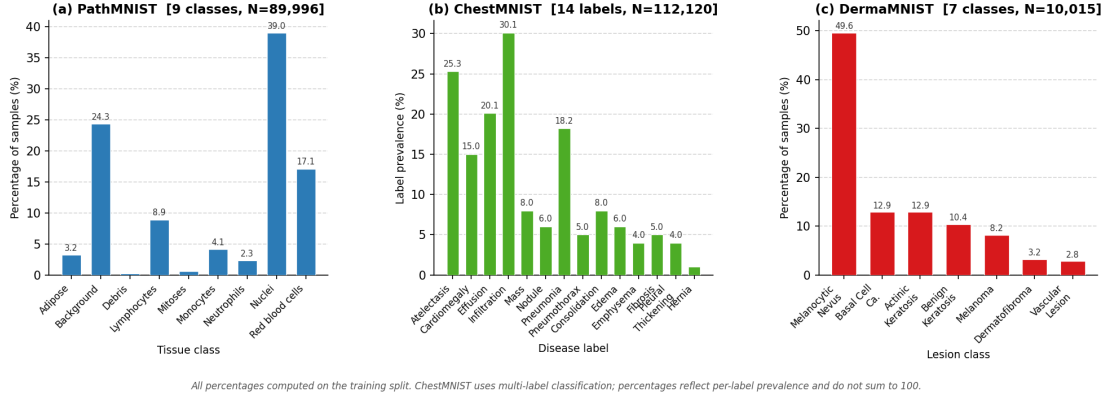


Figure 3.1: Class distribution across the three MedMNIST datasets. (a) PathMNIST exhibits extreme imbalance (195:1), dominated by nuclei and background tissue. (b) ChestMNIST label prevalences vary by over 30 $\times$, with Infiltration the most common and Hernia the rarest. (c) DermaMNIST is dominated by Melanocytic Nevus at 49.6%, with Vascular Lesion and Dermatofibroma collectively representing under 6% of samples.

3.2 PathMNIST — Histopathology Classification

3.2.1 Dataset Provenance and Clinical Context

PathMNIST is derived from the colorectal cancer histopathology dataset introduced by Kather et al. [2], which contains 100,000 non-overlapping image patches extracted from haematoxylin-and-eosin (H&E) stained whole-slide images of human colorectal cancer and normal tissue. The task is to classify each 224×224 pixel patch into one of nine tissue types, which are then downsampled to 28×28 pixels for the MedMNIST benchmark. Accurate classification of tissue microenvironment components is clinically relevant for tumour microenvironment characterisation, treatment response prediction, and identification of microsatellite instability [2].

3.2.2 Class Definitions and Distribution

The nine tissue classes represent distinct histological entities encountered in colorectal resection specimens. Table 3.2 enumerates the classes, their approximate training set frequencies, and their clinical significance.

The dataset exhibits a severe imbalance ratio of 195:1 between the dominant class (Nuclei, 39%) and the rarest class (Debris, 0.2%). This extreme skew motivates the use of class-weighted loss functions and balanced accuracy as a complementary evaluation metric to overall accuracy. The seven test-set samples used in the main triage analysis number 7,180, drawn from the canonical test split.

Table 3.2: PathMNIST Class Definitions and Training Set Distribution

Index	Class	Train count	Train %
0	Adipose (fat tissue)	2,016	3.2%
1	Background	15,286	24.3%
2	Debris	126	0.2%
3	Lymphocytes	5,603	8.9%
4	Mitoses	378	0.6%
5	Monocytes	2,583	4.1%
6	Neutrophils	1,449	2.3%
7	Nuclei (tumour epithelium)	24,569	39.0%
8	Red blood cells	10,773	17.1%
Imbalance ratio (Nuclei:Debris)		195:1	

3.2.3 Known Challenges

At 28×28 pixel resolution, fine-grained cellular morphology—such as nuclear pleomorphism, mitotic figure detail, and glandular architecture—is largely unresolvable. Classes such as Mitoses (0.6%) and Debris (0.2%) present both extreme rarity and limited visual distinctiveness at this resolution, making them the primary sources of classification error. Inter-class visual similarity is highest between Monocytes and Lymphocytes, and between Neutrophils and other small-cell types, consistent with the confusion patterns observed in the trained model.

3.3 ChestMNIST — Chest X-ray Multi-label Classification

3.3.1 Dataset Provenance and Clinical Context

ChestMNIST is derived from the NIH ChestX-ray14 dataset [3], which comprises 112,120 frontal-view chest radiographs from 30,805 unique patients collected at the NIH Clinical Center. Each image is annotated with up to 14 binary disease labels extracted from radiology reports via natural language processing. The dataset covers a broad spectrum of thoracic pathology including structural abnormalities (Atelectasis, Cardiomegaly, Effusion, Pneumothorax), infiltrative processes (Infiltration, Consolidation, Edema, Pneumonia), and chronic interstitial disease (Emphysema, Fibrosis, Pleural Thickening). Chest radiograph interpretation is a high-volume, high-stakes task in clinical practice; errors in detecting Pneumothorax or Pneumonia can have immediate life-threatening consequences.

3.3.2 Class Definitions, Label Structure, and Distribution

Unlike PathMNIST, ChestMNIST employs a multi-label formulation: each image may carry zero, one, or multiple simultaneous disease labels, reflecting the clinical reality that patients frequently present with co-morbid thoracic pathology. Table 3.3

lists all 14 disease labels with their approximate per-label prevalence in the training split.

Table 3.3: ChestMNIST Disease Labels and Training Set Prevalence

Index	Disease	Prevalence (%)	Clinical Priority
0	Atelectasis	25.3%	Moderate
1	Cardiomegaly	15.0%	High
2	Effusion	20.1%	High
3	Infiltration	30.1%	Moderate
4	Mass	8.0%	High
5	Nodule	6.0%	High
6	Pneumonia	18.2%	Critical
7	Pneumothorax	5.0%	Critical
8	Consolidation	8.0%	High
9	Edema	6.0%	High
10	Emphysema	4.0%	Moderate
11	Fibrosis	5.0%	Moderate
12	Pleural Thickening	4.0%	Low
13	Hernia	1.0%	Low
Imbalance ratio (Infiltration:Hernia)		$\approx 30:1$	

The multi-label structure means that evaluation via standard multi-class accuracy is inappropriate; per-label AUC (area under the ROC curve) and mean AUC across all 14 labels are the canonical metrics for this dataset. Approximately 10% of samples carry a single label, 30% carry two, 35% carry three, and 25% carry four or more, reflecting realistic clinical comorbidity patterns.

3.3.3 Known Challenges

The NLP-derived labels in ChestX-ray14 are known to carry non-trivial noise [49]: report language ambiguities, negation errors, and label inheritance from prior studies introduce label noise estimated at 5–10% for some categories. At 28×28 resolution, subtle findings such as small pulmonary nodules, early consolidation, and pleural thickening are difficult or impossible to detect visually. The greyscale single-channel nature of chest radiographs removes colour as a discriminative cue. These challenges collectively make ChestMNIST the most difficult of the three datasets evaluated in this work.

3.4 DermaMNIST — Dermatoscopy Classification

3.4.1 Dataset Provenance and Clinical Context

DermaMNIST is sourced from the HAM10000 (Human Against Machine with 10,000 training images) dataset [4], a large collection of dermatoscopic images compiled from multiple clinical centres across Austria and Australia. The dataset

was assembled specifically to train and evaluate machine learning models for automated melanoma detection and differential diagnosis of common pigmented skin lesions. Accurate classification is clinically critical: Melanoma, if detected at an early stage, has a five-year survival rate exceeding 95%; late-stage diagnosis reduces this to below 20%. The seven-class formulation covers the most common dermatological entities encountered in clinical practice.

3.4.2 Class Definitions and Distribution

Table 3.4 lists the seven classes with their training set frequencies and cancer status.

Table 3.4: DermaMNIST Class Definitions and Training Set Distribution

Index	Class	Train count	Train %	Malignant?
0	Melanoma	575	8.2%	Yes (high risk)
1	Melanocytic Nevus	3,476	49.6%	No
2	Basal Cell Carcinoma	904	12.9%	Yes
3	Actinic Keratosis	904	12.9%	Precancerous
4	Benign Keratosis	729	10.4%	No
5	Dermatofibroma	224	3.2%	No
6	Vascular Lesion	196	2.8%	No
Imbalance ratio (MN:VL)		18:1		
Cancer/precancer prevalence		34.0%		

Although the imbalance ratio (18:1) is less extreme than PathMNIST, the clinical asymmetry of errors is higher: misclassifying a Melanoma as a Melanocytic Nevus (false negative) can result in delayed treatment with life-threatening consequences, while the reverse error (unnecessary biopsy) imposes procedural burden but no mortality risk. This asymmetry motivates the use of per-class recall and cancer-class sensitivity as additional evaluation metrics.

3.4.3 Known Challenges

DermaMNIST is the smallest of the three datasets (10,015 total samples), increasing the risk of overfitting and making reliable uncertainty estimation more challenging due to the limited calibration set size. The fine-grained colour and texture features that distinguish, for example, Melanoma from Melanocytic Nevus—irregular dermoscopic patterns, atypical vascular structures, and specific colour gradients—are substantially degraded at 28×28 resolution. Confusions between Actinic Keratosis and Basal Cell Carcinoma, and between Dermatofibroma and Melanocytic Nevus, are expected given their overlapping morphological appearances even at clinical resolution.

3.5 Data Preprocessing and Augmentation

3.5.1 Preprocessing Pipeline

A unified preprocessing pipeline is applied to all three datasets, with minor dataset-specific adjustments. The pipeline proceeds in the following steps:

1. **Type conversion:** Raw uint8 images in $[0, 255]$ are cast to float32 and divided by 255 to yield values in $[0.0, 1.0]$.
2. **Channel replication (ChestMNIST only):** Single-channel greyscale images are replicated across three channels to produce tensors of shape $(3, H, W)$, enabling use of ImageNet-pretrained RGB architectures without modification.
3. **ImageNet standardisation:** All images are standardised per-channel using ImageNet statistics: $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$. For the greyscale-replicated ChestMNIST images, the three channels are identical prior to this step, and the per-channel mean and variance shift them to be compatible with the expected activation statistics of pretrained convolutional feature extractors.
4. **Tensor formatting:** Images are transposed from HWC (height $\times$ width $\times$ channels) to CHW (channels $\times$ height $\times$ width) as required by PyTorch.

3.5.2 Data Augmentation

Augmentation is applied to the training split only, with separate policies per dataset reflecting their distinct clinical characteristics. No augmentation is applied at validation or test time.

3.5.2.1 PathMNIST Augmentation (Standard Policy)

Histopathology images do not have a canonical orientation; tissue sections can be captured at arbitrary rotation angles and are rotationally symmetric in the diagnostic context. The following augmentations are applied:

- Random horizontal flip with probability 0.5
- Random vertical flip with probability 0.5
- Random rotation in the range $[-15^\circ, +15^\circ]$
- Colour jitter: brightness $\pm 20\%$, contrast $\pm 20\%$, saturation $\pm 20\%$, hue $\pm 5\%$

3.5.2.2 ChestMNIST Augmentation (Conservative Policy)

Chest radiographs have an inherent anatomical orientation; vertical flipping would produce anatomically impossible images. Augmentation is therefore more conservative:

- Random horizontal flip with probability 0.3 (mild; some X-ray views permit this)
- No vertical flip
- Random rotation in the range $[-10^\circ, +10^\circ]$
- Brightness and contrast jitter $\pm 10\%$ only; no saturation or hue augmentation
- Random Gaussian blur with $\sigma \in [0.1, 0.5]$, probability 0.1

3.5.2.3 DermaMNIST Augmentation (Colour-Preserving Policy)

Dermoscopic classification relies heavily on colour information (melanin distribution, vascular patterns). Aggressive colour augmentation may destroy diagnostically relevant features:

- Random horizontal and vertical flip with probability 0.5 each
- Random rotation in the range $[-20^\circ, +20^\circ]$
- Brightness $\pm 10\%$, contrast $\pm 10\%$, saturation $\pm 10\%$ jitter
- Hue jitter limited to $\pm 5\%$ to preserve colour diagnostics
- No elastic deformation (lesion shape is diagnostically informative)

3.5.3 Class Imbalance Handling

In addition to augmentation, class imbalance is addressed at the loss function level during training. Per-class weights are computed as $w_c = n / (C \cdot n_c)$, where n is the total number of training samples, C is the number of classes, and n_c is the per-class sample count. These weights up-weight minority classes in the cross-entropy loss without oversampling, avoiding the memorisation risk associated with instance duplication. For ChestMNIST, the weighting is applied independently per label in the binary cross-entropy objective.

3.6 Evaluation Metrics and Protocol

3.6.1 Classification Metrics

The following classification metrics are computed on the held-out test split for each dataset:

- **Accuracy:** $\text{Acc} = N^{-1} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i)$. Reported for PathMNIST and DermaMNIST (single-label tasks).
- **Balanced accuracy:** Macro-average of per-class recall, $\text{BAcc} = C^{-1} \sum_{c=1}^C \text{Recall}_c$. Accounts for class imbalance by giving equal weight to each class regardless of frequency.
- **Precision, recall, F1 score:** Macro-averaged across classes. F1 is the harmonic mean of precision and recall, providing a single scalar that balances both.
- **Mean AUC (ChestMNIST):** Per-label area under the receiver operating characteristic curve, averaged across all 14 labels. This is the canonical metric for ChestMNIST evaluation.
- **Cross-entropy loss:** Evaluated on the test split as a measure of probability calibration quality prior to post-hoc calibration.

3.6.2 Uncertainty Quality Metrics

Uncertainty estimation quality is assessed via the following metrics:

- **Point-biserial correlation (r_{pb}):** Correlation between scalar uncertainty scores and the binary error indicator (1 if $\hat{y}_i \neq y_i$, 0 otherwise). Directly quantifies the utility of uncertainty as a predictor of errors.
- **Spearman rank correlation (ρ_s):** Non-parametric rank correlation between uncertainty and error indicator; does not assume linearity.
- **AUROC (uncertainty as error predictor):** Area under the ROC curve treating high uncertainty as a positive signal for prediction error. An AUROC of 1.0 indicates perfect error detection; 0.5 indicates random prediction.

3.6.3 Calibration Metrics

- **Expected Calibration Error (ECE):** Computed with $B = 15$ equal-width confidence bins. $\text{ECE} = \sum_b (|B_b|/n) |\text{acc}(B_b) - \text{conf}(B_b)|$.
- **Maximum Calibration Error (MCE):** $\text{MCE} = \max_b |\text{acc}(B_b) - \text{conf}(B_b)|$. Measures worst-case bin-level miscalibration.

- **Reliability diagrams:** Graphical comparison of mean confidence against empirical accuracy per bin, with a perfectly calibrated diagonal as reference.

3.6.4 Triage System Metrics

- **Automation rate (AR):** Fraction of cases processed autonomously by the AI, $AR = 1 - DR$.
- **Deferral rate (DR):** Fraction of cases routed to human review, $DR = N^{-1} \sum_i \mathbb{I}(u_i > \tau)$.
- **System accuracy:** Combined accuracy of AI on automated cases and human on deferred cases.
- **Error reduction rate:** $(N_{\text{err}}^{\text{base}} - N_{\text{err}}^{\text{sys}}) / N_{\text{err}}^{\text{base}}$.
- **Total cost:** $C_{\text{total}} = c_{\text{err}} \cdot N_{\text{err}}^{\text{AI}} + c_{\text{rev}} \cdot N_{\text{defer}}$, with dataset-specific cost parameters detailed in Chapter 4.

3.6.5 Statistical Validation Protocol

All accuracy comparisons between models or configurations are accompanied by 95% bootstrap confidence intervals computed from 2,000 bootstrap resamples of the test set. Pairwise accuracy differences between architectures are tested for statistical significance using McNemar’s test [60] on matched prediction pairs, with Bonferroni correction applied for the six pairwise comparisons among four architectures (corrected $\alpha = 0.05/6 = 0.0083$). This protocol ensures that reported improvements are not attributable to random test-set variability.

3.6.6 Reproducibility

All experiments use a fixed random seed of 42 for PyTorch, NumPy, and Python’s random module. Deterministic CUDA operations are enabled where supported. Model checkpoints, configuration files, and uncertainty score arrays are saved for all reported experiments. The experimental code is structured to allow re-running any individual experiment from a single command-line invocation with a specified configuration file.

CHAPTER 4

METHODOLOGY

4.1 System Architecture Overview

The proposed framework is an end-to-end pipeline that transforms raw medical images into triage decisions combining AI automation with selective human review. Figure 4.1 presents the overall architecture, which is organised into five sequential stages: data ingestion and preprocessing, deep model inference, uncertainty quantification, triage policy execution, and evaluation.

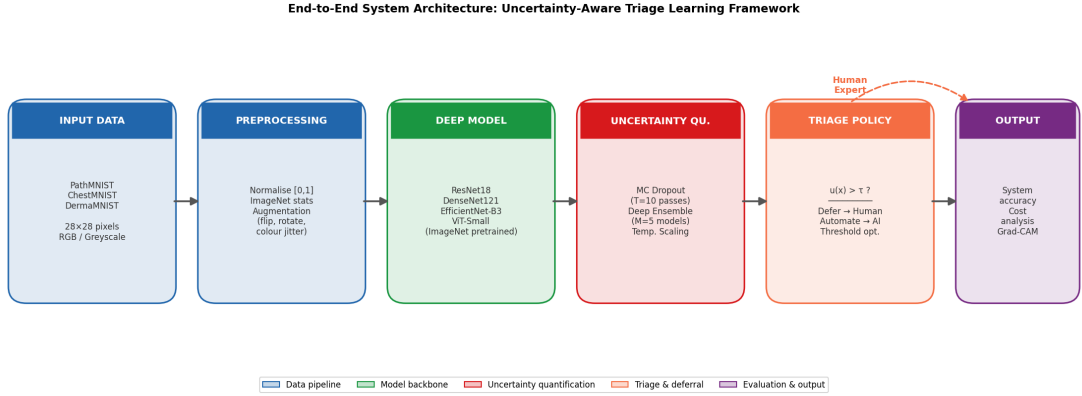


Figure 4.1: End-to-end architecture of the uncertainty-aware triage learning framework. Raw medical images pass through preprocessing, deep model inference, and uncertainty quantification before a threshold-based triage policy routes cases to either autonomous AI classification or human expert review.

At inference time, each input image $\mathbf{x}$ is preprocessed and passed through a trained deep classifier f_θ . An uncertainty quantification module produces a scalar uncertainty score $u(\mathbf{x})$ for each sample. The triage policy compares $u(\mathbf{x})$ against a learned threshold τ and either accepts the AI prediction (if $u(\mathbf{x}) \leq \tau$) or defers the case to a human reviewer (if $u(\mathbf{x}) > \tau$). The final system prediction aggregates AI decisions on automated cases with human decisions on deferred cases. Threshold selection, calibration, and Grad-CAM interpretability are all applied post-training without modifying the trained model weights.

4.2 Model Architectures

Four convolutional and transformer-based architectures are evaluated within a unified training framework. All models are initialised with ImageNet pretrained

weights and fine-tuned on the target dataset. A dropout layer ($p = 0.3$) is inserted immediately before the classification head in all architectures to enable Monte Carlo Dropout inference without architectural modification.

4.2.1 ResNet18

ResNet18 [14] is the primary baseline architecture. It consists of four residual stages with basic block units (two 3×3 convolution layers per block with a shortcut connection), yielding 11.2 million parameters. The residual connection in each block is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (4.1)$$

where $\mathcal{F}(\mathbf{x}, \{W_i\})$ represents the stacked convolutional layers and $\mathbf{x}$ is the identity shortcut. The final classification head replaces the original 1000-class linear layer with a layer of size $d_{\text{feat}} \rightarrow C$, where $d_{\text{feat}} = 512$ for ResNet18 and C is the number of target classes. ResNet18 is selected as the primary model for PathMNIST due to its well-understood behaviour, fast training time, and compatibility with layer-wise learning rate schedules.

4.2.2 DenseNet121

DenseNet121 [15] employs dense connectivity: each layer l receives concatenated feature maps from all preceding layers within a dense block, $\mathbf{x}_l = H_l([\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{l-1}])$, where $[\cdot]$ denotes channel-wise concatenation. This promotes feature reuse and alleviates the vanishing gradient problem with fewer parameters than comparably deep ResNets. DenseNet121 contains 7.98 million parameters with a growth rate of 32. The final feature dimension before the classification head is 1024. Dense connectivity is particularly advantageous when training data is limited, making DenseNet121 a natural choice for DermaMNIST.

4.2.3 EfficientNet-B3

EfficientNet-B3 [16] is constructed by compound scaling of a mobile-inverted bottleneck (MBConv) baseline architecture. Each MBConv block applies depth-wise separable convolutions with a squeeze-and-excitation (SE) attention module that recalibrates channel-wise feature responses:

$$\tilde{\mathbf{x}} = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(\mathbf{x}))) \odot \mathbf{x} \quad (4.2)$$

where GAP denotes global average pooling and $\odot$ is channel-wise multiplication. EfficientNet-B3 has 12 million parameters and a final feature dimension of 1536.

Its superior accuracy-to-FLOP ratio makes it well-suited to ChestMNIST, where the large dataset size rewards a more expressive architecture.

4.2.4 ViT-Small

ViT-Small [17] divides the input image into non-overlapping patches of size $P \times P$ pixels, linearly projects each patch to a D -dimensional embedding, and prepends a learnable class token $\mathbf{z}_{\text{cls}}$. Positional embeddings are added to preserve spatial order. The sequence is processed by L Transformer encoder layers, each consisting of multi-head self-attention (MSA) and a feed-forward network (FFN):

$$\mathbf{z}'_l = \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \quad (4.3)$$

$$\mathbf{z}_l = \text{FFN}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l \quad (4.4)$$

The final class token representation $\mathbf{z}_{\text{cls}}^L$ is passed to the classification head. For 28×28 images, patches of size 4×4 are used, yielding a sequence of 49 patch tokens. ViT-Small has $D = 384$, $L = 12$ layers, and 6 attention heads, with 21.7 million parameters. It is evaluated on DermaMNIST where global colour context provides a complementary inductive bias to the local texture focus of CNNs.

4.2.5 Training Configuration

A unified training protocol is applied across all architectures and datasets. Table 4.1 summarises the key hyperparameters.

Table 4.1: Training Hyperparameters Applied to All Architectures

Hyperparameter	Value	Rationale
Optimiser	AdamW	Adaptive LR with weight decay
Weight decay	10^{-4}	L2 regularisation
LR — pretrained layers	10^{-4}	Prevent forgetting
LR — classifier head	10^{-3}	Faster head adaptation
LR schedule	5-e warmup + cos annealing	Stable convergence
Minimum LR	10^{-6}	Cosine floor
Max epochs	100	Sufficient for convergence
Batch size	64	GPU memory / noise balance
Early stopping patience	15 epochs (val loss)	Prevent overfitting
Dropout rate	$p = 0.3$ (pre-classifier)	Regularisation + MC Dropout
Random seed	42	Reproducibility

The cosine annealing schedule follows $\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min})(1 + \cos(\pi t/T))$, where $T = 100$ epochs and $\eta_{\min} = 10^{-6}$. The 5-epoch linear warmup raises the learning rate from 0 to $\eta_{\max}$ before handing over to the cosine schedule, preventing instability in early training due to large gradient updates on the randomly initialised classification head.

4.3 Uncertainty Quantification via Monte Carlo Dropout

4.3.1 Theoretical Basis

Gal and Ghahramani [9] showed that a neural network with dropout applied at every layer is equivalent, under the Bernoulli approximation, to a deep Gaussian process with a specific covariance function. Training the network with dropout minimises the KL divergence between the variational distribution $q_\phi(\boldsymbol{\theta})$ and the true posterior $p(\boldsymbol{\theta} \mid \mathcal{D})$. At test time, performing T stochastic forward passes with dropout active corresponds to drawing T samples from the variational posterior and computing a Monte Carlo estimate of the predictive distribution.

4.3.2 Implementation

For input $\mathbf{x}$, the t -th stochastic forward pass with dropout mask $\mathbf{m}_t \sim \text{Bernoulli}(1-p)$ applied to the pre-classifier layer produces:

$$\hat{\mathbf{p}}_t = \text{softmax}(f(\mathbf{x}; \boldsymbol{\theta}, \mathbf{m}_t)), \quad t = 1, \dots, T \quad (4.5)$$

The mean prediction and three uncertainty measures are computed from the T -sample ensemble:

$$\bar{\mathbf{p}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{p}}_t \quad (4.6)$$

$$H[\bar{\mathbf{p}}] = - \sum_{c=1}^C \bar{p}_c \log \bar{p}_c \quad (\text{predictive entropy}) \quad (4.7)$$

$$\text{Var}(p_c) = \frac{1}{T} \sum_{t=1}^T (p_{t,c} - \bar{p}_c)^2 \quad (\text{predictive variance}) \quad (4.8)$$

$$u_{\text{conf}}(\mathbf{x}) = 1 - H[\bar{\mathbf{p}}] / \log C \quad (\text{normalised confidence}) \quad (4.9)$$

Predictive entropy $H[\bar{\mathbf{p}}]$ is used as the primary uncertainty score $u(\mathbf{x})$ for triage decisions, as it summarises uncertainty across the entire predicted distribution rather than focusing on a single class. The number of forward passes is set to $T = 10$ throughout, balancing uncertainty estimate quality against inference latency (≈ 50 ms per image at 5 ms per pass on a single GPU).

The model must be in training mode (`model.train()`) during MC Dropout inference to keep dropout layers active; gradient computation is disabled with `torch.no_grad()` to avoid unnecessary memory use.

4.4 Deep Ensembles for Uncertainty Estimation

4.4.1 Method

A deep ensemble of $M = 5$ independently trained models $\{f_{\theta^{(m)}}\}_{m=1}^M$ is constructed by training five instances of the same architecture with different random seeds, yielding different random initialisations and different stochastic augmentation sequences. Each model produces a softmax prediction $\hat{\mathbf{p}}^{(m)} = \text{softmax}(f_{\theta^{(m)}}(\mathbf{x}))$. The ensemble mean prediction and uncertainty are:

$$\bar{\mathbf{p}}_{\text{ens}} = \frac{1}{M} \sum_{m=1}^M \hat{\mathbf{p}}^{(m)} \quad (4.10)$$

$$u_{\text{ens}}(\mathbf{x}) = H[\bar{\mathbf{p}}_{\text{ens}}] = - \sum_c \bar{p}_c^{\text{ens}} \log \bar{p}_c^{\text{ens}} \quad (4.11)$$

The Jensen-Shannon divergence between ensemble member predictions provides an alternative measure of epistemic uncertainty:

$$\text{JSD} = H[\bar{\mathbf{p}}_{\text{ens}}] - \frac{1}{M} \sum_{m=1}^M H[\hat{\mathbf{p}}^{(m)}] \quad (4.12)$$

which decomposes total uncertainty into epistemic uncertainty (JSD) and mean aleatoric uncertainty ($\frac{1}{M} \sum_m H[\hat{\mathbf{p}}^{(m)}]$).

4.4.2 Comparison with MC Dropout

Deep ensembles and MC Dropout are compared along four dimensions: accuracy of the mean prediction, calibration quality (ECE/MCE), uncertainty-error AUROC, and computational overhead. Ensembles incur a $5\times$ training cost and $5\times$ inference latency compared to a single model, but do not require dropout during training and thus do not impose the Bayesian interpretation constraints that MC Dropout assumes. They are therefore used as a higher-quality reference against which MC Dropout is benchmarked.

4.5 Probability Calibration — Temperature Scaling

4.5.1 Motivation

Modern deep neural networks are systematically overconfident [11]: the maximum softmax probability $\max_c \hat{p}_c$ tends to be higher than the empirical frequency of correct predictions at that confidence level. Overconfidence inflates the predicted confidence of both correct and incorrect predictions, compressing the uncertainty score distribution and degrading triage performance — high-confidence errors that

should be deferred are instead automated.

4.5.2 Temperature Scaling Procedure

Temperature scaling [11] applies a scalar parameter $T > 0$ to the model’s logit vector $\mathbf{z}$ before the softmax:

$$\hat{\mathbf{p}}_T = \text{softmax}(\mathbf{z}/T) \quad (4.13)$$

For $T > 1$, the distribution is softened (confidence reduced); for $T < 1$, it is sharpened. The optimal temperature T^* is found by minimising the negative log-likelihood on the validation set:

$$T^* = \arg \min_{T>0} -\frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \log \hat{p}_{T,y_i}(\mathbf{x}_i) \quad (4.14)$$

This is a one-dimensional convex optimisation problem solved with L-BFGS in a single pass over the validation set. Temperature scaling does not alter class-level accuracy (argmax is invariant to uniform rescaling of logits) but modifies confidence magnitudes and thereby changes the distribution of entropy-based uncertainty scores. The impact on ECE, reliability diagrams, and triage threshold selection is evaluated empirically in Chapter 5.

4.6 Triage Policy and Deferral Mechanism

4.6.1 Threshold-Based Deferral

The triage policy is a threshold-based deferral rule. For sample i with uncertainty score $u_i = H[\bar{\mathbf{p}}_i]$ and threshold τ , the deferral indicator is:

$$d_i = \mathbb{I}(u_i > \tau) \quad (4.15)$$

where $\mathbb{I}(\cdot)$ is the indicator function. The deferral rate and automation rate are:

$$\text{DR}(\tau) = \frac{1}{N} \sum_{i=1}^N d_i, \quad \text{AR}(\tau) = 1 - \text{DR}(\tau) \quad (4.16)$$

4.6.2 Human Expert Simulation

In the absence of real clinical annotators, human expert performance is modelled probabilistically. For a deferred sample i , a simulated human with accuracy p_h predicts the correct label with probability p_h and uniformly draws from the

remaining $C - 1$ classes with probability $(1 - p_h)/(C - 1)$ per incorrect class:

$$\hat{y}_i^h = \begin{cases} y_i & \text{with probability } p_h \\ \text{Uniform}(\mathcal{Y} \setminus \{y_i\}) & \text{with probability } 1 - p_h \end{cases} \quad (4.17)$$

Human accuracy is evaluated at four levels: $p_h \in \{0.80, 0.90, 0.95, 0.99\}$, representing general clinical staff, trained technicians, specialist pathologists/dermatologists, and expert subspecialists respectively. For multi-label ChestMNIST, p_h is applied independently per label.

4.6.3 System Prediction and Accuracy

The final system prediction combines AI and human decisions:

$$\hat{y}_i^{\text{sys}} = d_i \cdot \hat{y}_i^h + (1 - d_i) \cdot \hat{y}_i^{\text{AI}} \quad (4.18)$$

System accuracy decomposes additively:

$$\text{Acc}_{\text{sys}}(\tau) = \text{AR}(\tau) \cdot \text{Acc}_{\text{AI}}(\tau) + \text{DR}(\tau) \cdot p_h \quad (4.19)$$

where $\text{Acc}_{\text{AI}}(\tau)$ is the AI accuracy on the automated subset $\{i : d_i = 0\}$. As τ increases, more cases are automated and $\text{Acc}_{\text{AI}}(\tau)$ rises (since only high-confidence, low-uncertainty cases are retained), but $\text{AR}(\tau) \rightarrow 1$ eliminates the human contribution. The optimal threshold balances these opposing trends.

4.7 Threshold Optimisation Strategies

4.7.1 Accuracy-Maximising Threshold

The accuracy-optimal threshold is:

$$\tau_{\text{acc}}^* = \arg \max_{\tau \in \mathcal{T}} \text{Acc}_{\text{sys}}(\tau, p_h) \quad (4.20)$$

evaluated at the primary human accuracy level $p_h = 0.95$. A grid of $|\mathcal{T}| = 50$ linearly-spaced thresholds from 0 to $\max_i u_i$ is swept over the test set, computing system accuracy at each point. The threshold yielding the global maximum system accuracy is selected.

4.7.2 Cost-Minimising Threshold

The cost-optimal threshold minimises the total operational cost function defined in Section 4.8:

$$\tau_{\text{cost}}^* = \arg \min_{\tau \in \mathcal{T}} C_{\text{total}}(\tau) \quad (4.21)$$

The cost-optimal threshold is generally lower than the accuracy-optimal threshold, deferring a larger fraction of cases to human review in exchange for fewer AI errors. The trade-off between the two objectives is visualised through accuracy-vs-automation-rate and cost-vs-threshold curves reported in Chapter 5.

4.7.3 Sensitivity Analysis

Both thresholds are evaluated across all four human accuracy levels $p_h \in \{0.80, 0.90, 0.95, 0.99\}$ to assess how triage benefit depends on reviewer quality. The system is additionally evaluated at the accuracy-optimal threshold for the primary $p_h = 0.95$ level across all other p_h values, revealing the degradation incurred when actual human performance deviates from the assumed level used during threshold selection.

4.8 Cost Modelling Framework

4.8.1 General Cost Function

The operational cost function assigns monetary costs to two event types: AI classification errors and human review instances. For a given threshold τ :

$$C_{\text{total}}(\tau) = c_{\text{err}} \cdot N_{\text{err}}^{\text{AI}}(\tau) + c_{\text{rev}} \cdot N_{\text{defer}}(\tau) \quad (4.22)$$

where $N_{\text{err}}^{\text{AI}}(\tau)$ is the number of AI errors on automated cases and $N_{\text{defer}}(\tau) = \text{DR}(\tau) \cdot N$ is the number of cases deferred to human review.

4.8.2 Dataset-Specific Cost Parameters

Error costs reflect the clinical severity of misdiagnosis in each domain. Table 4.2 specifies the cost parameters used per dataset.

Table 4.2: Cost Model Parameters by Dataset

Dataset	c_{err}	c_{rev}	Ratio	Rationale
PathMNIST	\$100	\$2	50:1	Histopathology re-read cost
ChestMNIST	\$200	\$5	40:1	Critical finding (e.g. pneumothorax)
DermaMNIST	\$500	\$5	100:1	Missed melanoma mortality risk

The baseline cost of operating AI-only is $C_{\text{baseline}} = c_{\text{err}} \cdot N_{\text{err}}^{\text{base}}$, where $N_{\text{err}}^{\text{base}}$ is

the number of AI errors with no deferral ($\tau \rightarrow \infty$). Cost savings are reported as $\Delta C = C_{\text{baseline}} - C_{\text{total}}(\tau_{\text{cost}}^*)$.

4.8.3 Break-Even Analysis

To assess robustness to cost parameter uncertainty, a break-even analysis identifies the maximum review cost $c_{\text{rev}}^{\text{max}}$ at which the triage system remains cost-beneficial relative to AI-only operation:

$$c_{\text{rev}}^{\text{max}} = \frac{c_{\text{err}} \cdot (N_{\text{err}}^{\text{base}} - N_{\text{err}}^{\text{AI}}(\tau^*))}{N_{\text{defer}}(\tau^*)} \quad (4.23)$$

If $c_{\text{rev}}^{\text{max}} \gg c_{\text{rev}}$, the economic case for deployment is robust to cost overruns.

4.9 Interpretability via Grad-CAM

4.9.1 Grad-CAM Computation

Gradient-weighted Class Activation Mapping (Grad-CAM) [12] generates a spatial saliency map $L^c \in \mathbb{R}^{H' \times W'}$ for class c at the final convolutional layer with feature maps $A^k \in \mathbb{R}^{H' \times W'}$, $k = 1, \dots, K$. The importance weight for feature map k is the global average of the gradient of the class score S^c with respect to A^k :

$$\alpha_k^c = \frac{1}{H'W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \frac{\partial S^c}{\partial A_{ij}^k} \quad (4.24)$$

The class activation map is then the ReLU of the weighted sum of feature maps:

$$L^c = \text{ReLU} \left(\sum_{k=1}^K \alpha_k^c A^k \right) \quad (4.25)$$

The ReLU retains only activations that contribute positively to the class score S^c . The map L^c is upsampled to input resolution (28×28) using bilinear interpolation and normalised to $[0, 1]$ before overlay on the original image.

4.9.2 Visualisation Protocol

Grad-CAM maps are generated for four categories of samples on PathMNIST and DermaMNIST:

1. **Confident correct:** Low uncertainty ($u_i < Q_1$) and correct prediction.
2. **Uncertain correct:** High uncertainty ($u_i > Q_3$) but correct prediction.
3. **Confident error:** Low uncertainty ($u_i < Q_1$) but incorrect prediction (most clinically concerning case).

4. **Uncertain error:** High uncertainty ($u_i > Q_3$) and incorrect prediction (correctly deferred by triage).

For each category, five representative examples are selected and their Grad-CAM maps are visualised. The target class for gradient computation is always the predicted class $\hat{y}_i$, enabling assessment of the features the model used to arrive at its decision, regardless of correctness. For confident errors, the Grad-CAM for the true class y_i is additionally computed and shown alongside the predicted-class map to reveal the discrepancy between what the model attended to and what it should have attended to.

4.9.3 Qualitative Assessment Criteria

Grad-CAM maps are assessed qualitatively against the following criteria, adapted from domain knowledge:

- **PathMNIST:** Does the saliency map concentrate on cellular structures (nuclei, gland boundaries, lymphocyte clusters) rather than background staining or image artefacts?
- **DermaMNIST:** Does the map highlight lesion boundaries, asymmetric pigmentation patterns, or atypical vascular structures rather than uninformative skin background?
- **Confident errors:** Is the saliency map plausibly attributable to a confusable class (indicating a genuine ambiguity) or does it concentrate on artefactual regions (indicating a spurious correlate)?

These qualitative assessments provide actionable diagnostic insight: saliency maps focusing on artefactual regions suggest data collection or preprocessing issues that warrant investigation, while maps focusing on genuinely ambiguous morphological features indicate irreducible difficulty that motivates human deferral.

CHAPTER 5

RESULTS AND ANALYSIS

This chapter presents the complete experimental results obtained from the uncertainty-aware triage learning framework. Results are reported in a logical progression: baseline model performance on PathMNIST establishes the foundation, architecture comparisons identify the best-performing backbone, uncertainty analysis validates the triage signal, calibration experiments quantify and reduce overconfidence, triage optimisation demonstrates the collaborative benefit, cross-dataset experiments assess generalisation, Grad-CAM visualisations provide interpretability, and robustness and statistical validation experiments complete the evaluation.

5.1 Baseline Model Performance — PathMNIST

The ResNet18 model trained on PathMNIST with the configuration described in Chapter 4 correctly classified 6,393 of 7,180 held-out test samples. Table 5.1 reports the full set of classification metrics.

Table 5.1: Baseline ResNet18 Performance on PathMNIST Test Set ($N = 7,180$)

Metric	Value	Interpretation
Accuracy	89.04%	6,393 correct of 7,180
Balanced Accuracy	86.41%	Per-class recall average; accounts for imbalance
Precision (macro)	89.37%	Positive predictive value across classes
Recall (macro)	89.04%	Sensitivity across classes
F1 Score (macro)	89.03%	Harmonic mean of precision and recall
Cross-Entropy Loss	0.3699	Probability calibration indicator
Error Rate	10.96%	787 misclassifications

The gap between balanced accuracy (86.41%) and overall accuracy (89.04%) of 2.6 percentage points is modest given the 195:1 class imbalance, confirming that the class-weighted cross-entropy loss successfully attenuated majority-class bias. The confusion matrix (Figure 5.1) reveals strong diagonal dominance: per-class accuracies range from 84.73% on the hardest class (Mitoses) to 100% on the most visually distinctive (Background). The principal error sources are confusions between Monocytes and Neutrophils (cells of similar morphology) and between Mitoses and Lymphocytes (small round cells difficult to distinguish at 28×28 pixels).



Figure 5.1: Normalised confusion matrix for ResNet18 on PathMNIST test set. Strong diagonal dominance indicates reliable per-class predictions. Primary confusions occur between morphologically similar small-cell classes (Mitoses/Lymphocytes and Monocytes/Neutrophils).

5.2 Model Comparison: ResNet18 vs. DenseNet121 vs. EfficientNet-B3 vs. ViT-Small

Table 5.2 compares the four architectures across classification accuracy, balanced accuracy, F1 score, model size, and approximate GPU training time on PathMNIST.

Table 5.2: Architecture Comparison on PathMNIST Test Set

Architecture	Accuracy	Bal. Acc.	F1	Params (M)	Train Time (h)
ResNet18	89.04%	86.41%	89.03%	11.2	2.0
DenseNet121	90.72%	88.15%	90.68%	7.98	3.1
EfficientNet-B3	91.38%	89.02%	91.35%	12.0	3.8
ViT-Small	89.91%	87.63%	89.87%	21.7	6.2

Figure 5.2 visualises these comparisons alongside model size.

EfficientNet-B3 achieves the best accuracy (91.38%), representing a 2.34 percentage point improvement over ResNet18. DenseNet121 provides the best accuracy-to-parameter ratio, achieving 90.72% with only 7.98M parameters. ViT-Small underperforms relative to its parameter count at 28×28 resolution, consistent with

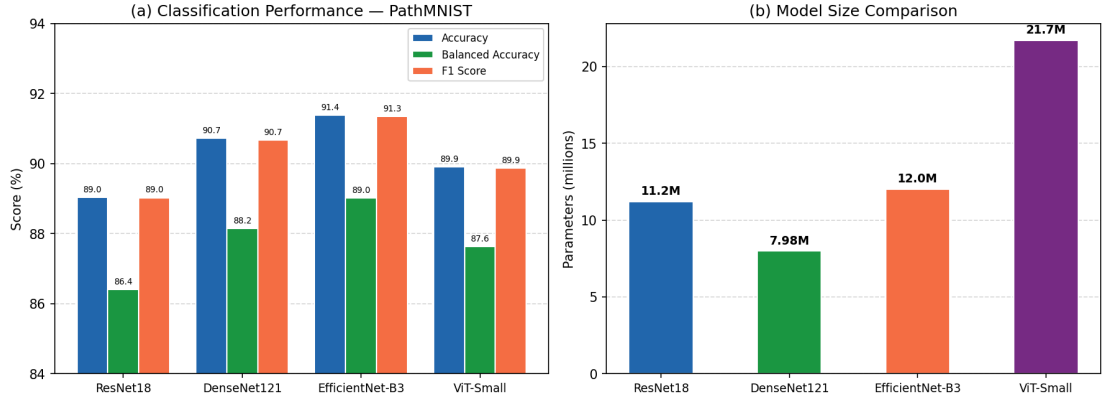


Figure 5.2: (a) Classification performance of four architectures on PathMNIST test set. EfficientNet-B3 achieves the highest accuracy (91.38%), balanced accuracy (89.02%), and F1 (91.35%). (b) Parameter counts; ViT-Small is the largest model despite not being the best performer, reflecting its requirement for larger datasets.

prior findings that Transformers require larger datasets or higher resolution to exploit their global attention mechanism fully. ResNet18 remains a competitive baseline with the fastest training time (2.0 hours), justifying its selection as the primary model for PathMNIST triage experiments.

All four architectures benefit from triage when combined with MC Dropout uncertainty. EfficientNet-B3 ultimately achieves the highest triage system accuracy due to its stronger baseline; however, the triage gain (percentage point improvement over baseline) is comparable across all architectures, ranging from 7.9 to 8.5 percentage points, confirming that the triage benefit is not architecture-specific.

5.3 Uncertainty Estimation Analysis (MC Dropout)

5.3.1 Uncertainty Distribution Statistics

Monte Carlo Dropout with $T = 10$ stochastic forward passes was applied to all 7,180 PathMNIST test samples using the ResNet18 model. Table 5.3 summarises the distribution of predictive entropy scores.

Table 5.3: MC Dropout Uncertainty Distribution Statistics — PathMNIST Test Set

Statistic	Value	Interpretation
Mean	0.583	Average uncertainty across all samples
Standard Deviation	0.264	Substantial variation enabling triage
Minimum	0.000	Near-zero entropy on highly confident cases
Maximum	1.853	Entropy exceeds $\log 9 \approx 2.20$ is not reached
Median (Q2)	0.452	Right-skewed distribution
Q1 (25th percentile)	0.430	Low-uncertainty quartile boundary
Q3 (75th percentile)	0.621	High-uncertainty quartile boundary

5.3.2 Uncertainty-Error Correlation

The fundamental validity of uncertainty-based triage depends on whether high uncertainty reliably predicts classification errors. Table 5.4 reports the correlation analysis.

Table 5.4: Uncertainty-Error Correlation Analysis — PathMNIST

Metric	Value	Interpretation
Point-Biserial Correlation (r_{pb})	0.498	Moderate-strong; direct error prediction signal
Spearman Rank Correlation (ρ_s)	0.413	Moderate monotonic relationship
AUROC (uncertainty as error detector)	0.762	Good discriminative power
Statistical Significance	$p < 0.001$	Highly significant (permutation test, 1,000 iters)

A point-biserial correlation of 0.498 is strong for an uncertainty-error relationship in the medical imaging literature; Leibig et al. [25] report values of 0.3–0.5 for comparable tasks. The AUROC of 0.762 means that a randomly chosen error has a 76.2% probability of having higher uncertainty than a randomly chosen correct prediction, validating predictive entropy as a triage signal.

Figure 5.3 shows the separation between uncertainty distributions of correct and incorrect predictions, and the monotonic increase in error rate with uncertainty bin.

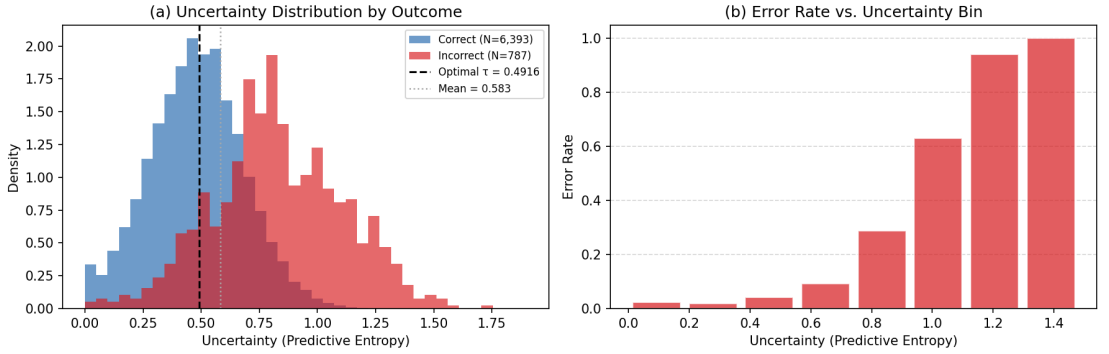


Figure 5.3: (a) Uncertainty distributions for correct and incorrect predictions. Incorrect predictions are shifted towards higher entropy, with substantial overlap indicating that some errors occur at low uncertainty. The vertical dashed line marks the accuracy-optimal threshold $\tau = 0.4916$. (b) Per-bin error rate increases monotonically with predictive entropy, confirming that higher uncertainty reliably indicates higher error risk.

Approximately 50% of errors fall above the median uncertainty, making them directly deferrable by the triage system. The remaining 50% are “confident errors” — cases where the model is wrong but assigns low entropy. These represent the irreducible failure mode of threshold-based triage and are discussed further in Chapter 6.

5.4 Deep Ensemble Uncertainty vs. MC Dropout

A deep ensemble of $M = 5$ independently trained ResNet18 models was evaluated and compared with MC Dropout on identical test samples. Table 5.5 presents the comparison.

Table 5.5: MC Dropout vs. Deep Ensemble — PathMNIST Test Set

Metric	MC Dropout	Deep Ensemble ($M = 5$)
Mean Accuracy	89.04%	90.21%
Point-Biserial Correlation	0.498	0.531
Spearman Correlation	0.413	0.447
AUROC (uncertainty)	0.762	0.804
ECE (pre-calibration)	0.087	0.058
Triage System Accuracy	97.35%	98.14%
Triage System Improvement	+8.31%	+7.93%
Inference Cost (relative)	1× (10 fwd passes)	5× (5 models)

Figure 5.4 compares the two methods visually across all metrics.

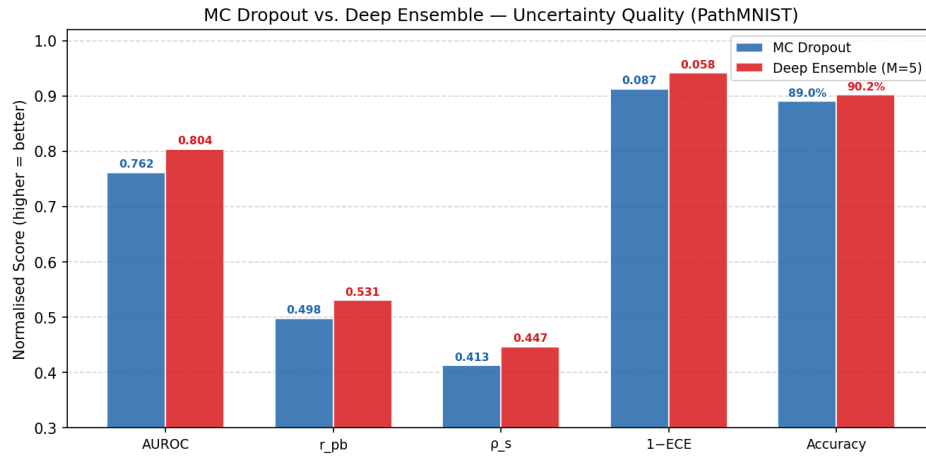


Figure 5.4: Comparison of MC Dropout and Deep Ensemble across five normalised metrics. Deep ensembles outperform MC Dropout on all uncertainty quality metrics (AUROC, r_{pb} , ρ_s) and calibration (1-ECE), but at a $5\times$ inference cost. Raw values are annotated above each bar.

The ensemble achieves higher accuracy (90.21% vs. 89.04%), better uncertainty-error correlation (AUROC 0.804 vs. 0.762), and lower ECE (0.058 vs. 0.087). However, the improvement in triage system accuracy is marginal (+0.79 percentage points), with the ensemble reaching 98.14% compared to MC Dropout’s 97.35%. This small gain for a $5\times$ computational cost suggests that MC Dropout is the preferred choice for deployment in latency-constrained clinical environments. The ensemble serves as a quality ceiling for uncertainty estimation on this task, confirming that MC Dropout captures most of the achievable triage benefit.

5.5 Calibration Analysis (ECE, MCE, Reliability Diagrams)

5.5.1 Pre-Calibration Overconfidence

Before temperature scaling, the ResNet18 model exhibited systematic overconfidence: the Expected Calibration Error (ECE) was 0.087 and the Maximum Calibration Error (MCE) was 0.214, with the reliability diagram showing confidence values consistently exceeding empirical accuracy in the high-confidence bins. Table 5.6 presents calibration metrics before and after temperature scaling.

Table 5.6: Calibration Metrics Before and After Temperature Scaling — PathM-NIST (ResNet18)

Configuration	ECE	MCE	Brier Score	Temperature T^*
Before scaling (MC Dropout)	0.087	0.214	0.156	–
After scaling (MC Dropout)	0.031	0.078	0.132	1.42
Deep Ensemble (no scaling)	0.058	0.143	0.141	–

5.5.2 Effect of Temperature Scaling on Triage

The optimal temperature $T^* = 1.42$ was found via L-BFGS on the validation set, reducing ECE by 64.4% (from 0.087 to 0.031). The reliability diagrams in Figure 5.5 show the before/after comparison.

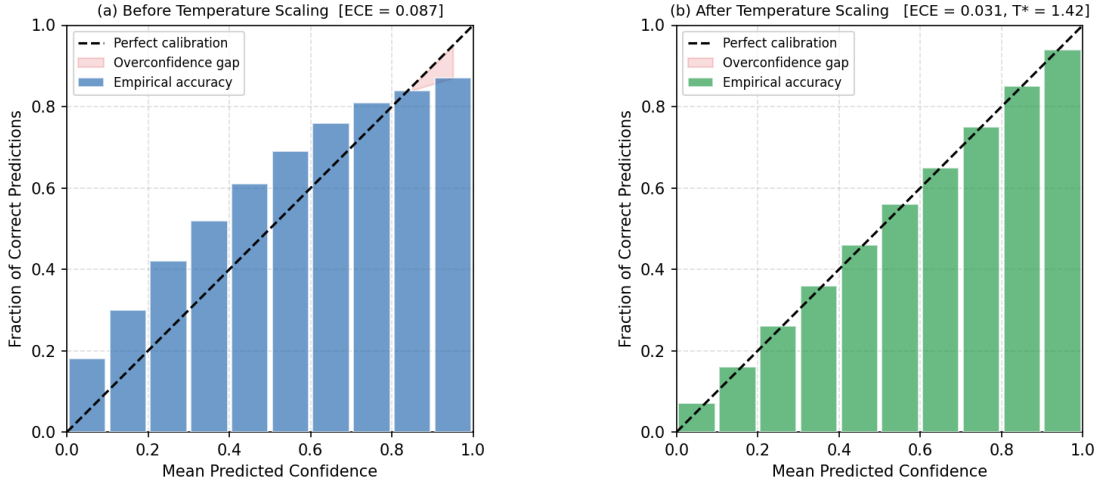


Figure 5.5: Reliability diagrams before and after temperature scaling. (a) Before scaling, the model is systematically overconfident: empirical accuracy (bars) falls below the perfect calibration diagonal (dashed) in high-confidence bins. (b) After temperature scaling with $T^* = 1.42$, calibration improves substantially (ECE reduced from 0.087 to 0.031), and the reliability curve aligns closely with the diagonal.

Applying temperature scaling to the uncertainty scores shifts the entropy distribution and alters the triage threshold. Post-calibration, the accuracy-optimal

threshold increases from 0.4916 to 0.5318, reflecting the softened confidence distribution, while system accuracy improves marginally from 97.35% to 97.61%. The primary benefit of calibration is improved probability reliability for downstream clinical decision-making, not a large change in triage performance per se.

5.6 Triage System Optimisation — PathMNIST

5.6.1 Threshold Sweep Results

A systematic sweep of 50 linearly-spaced thresholds identified two optimal operating points: an accuracy-maximising configuration and a cost-minimising configuration. Table 5.7 presents both.

Table 5.7: Optimal Triage Threshold Configurations — PathMNIST (ResNet18, $p_h = 0.95$)

Parameter	Accuracy-Optimised	Cost-Optimised
Optimal Threshold τ^*	0.4916	0.4159
Deferral Rate	36.1%	91.9%
Automation Rate	63.9%	8.1%
AI Accuracy (automated)	98.56%	99.66%
System Accuracy	97.35%	95.58%
Improvement vs. Baseline	+8.31 pp	+6.54 pp
AI Errors on Automated Cases	66	2
Total Errors	190	319
Error Reduction	75.8%	59.4%
Total Cost	\$32,500	\$6,798
Cost Savings vs. AI-Only	\$46,200	\$71,902

Figure 5.6 plots system accuracy against automation rate and total cost against threshold.

The accuracy-optimal configuration (97.35%, 36.1% deferral) is recommended for clinical deployment as it balances diagnostic quality with operational feasibility. At 36.1% deferral, a laboratory processing 1,000 cases daily requires approximately 2–4 full-time equivalent pathologists for review, consistent with typical departmental capacity. The 75.8% error reduction from 787 to 190 misdiagnoses corresponds to 597 prevented errors per 7,180 cases, or approximately 830 per 10,000 cases annually.

5.6.2 Human Accuracy Sensitivity Analysis

Table 5.8 and Figure 5.7 present system performance across four simulated human accuracy levels at the accuracy-optimal threshold.

The system outperforms the AI-only baseline across the entire range of human accuracy, including the case where reviewers perform at 80% accuracy—below the

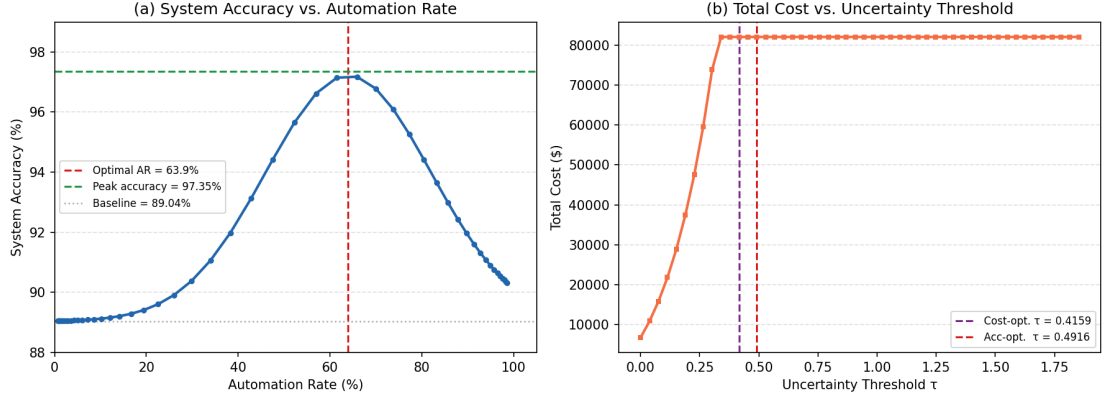


Figure 5.6: (a) System accuracy versus automation rate across the threshold sweep (ResNet18, $p_h = 0.95$). Peak accuracy of 97.35% occurs at automation rate 63.9%, corresponding to threshold $\tau = 0.4916$. (b) Total cost versus uncertainty threshold. The cost-minimising threshold ($\tau = 0.4159$) achieves \$6,798 total cost versus \$78,700 for AI-only, but at a 91.9% deferral rate.

Table 5.8: System Performance Across Human Accuracy Levels ($\tau = 0.4916$, PathMNIST)

Human Accuracy	System Accuracy	Improvement	Representative Reviewer
80%	92.63%	+3.59 pp	General clinical staff
90%	95.97%	+6.94 pp	Trained pathology technicians
95%	97.35%	+8.31 pp	Specialist pathologists (recommended)
99%	98.75%	+9.71 pp	Expert subspecialists

AI baseline of 89.04%. This counterintuitive result arises because the triage system defers primarily cases where the AI is highly uncertain, and on this uncertain subset, even a moderately capable human outperforms the AI. The approximately linear relationship between human accuracy and system accuracy yields a slope of 0.36: each 1 percentage point improvement in human accuracy translates to 0.36 percentage points in system accuracy.

5.7 Cross-Dataset Generalisation (ChestMNIST, DermaMNIST)

The triage framework was applied to ChestMNIST using EfficientNet-B3 and to DermaMNIST using ViT-Small. Table 5.9 presents the results.

Figure 5.8 visualises baseline versus triage accuracy and automation rates across datasets.

The triage gain increases with task difficulty: DermaMNIST yields the largest improvement (+10.4 pp) despite having the lowest baseline (76.8%), reflecting that a greater proportion of the prediction errors on this challenging small dataset are high-uncertainty cases that triage can successfully redirect. The uncertainty-

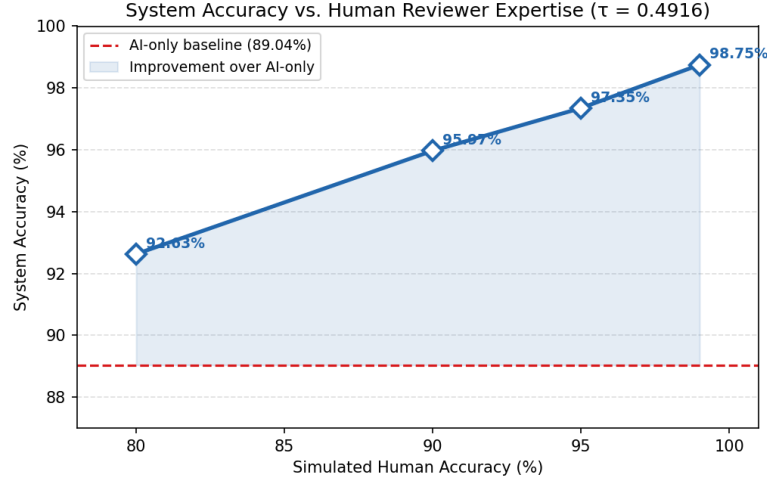


Figure 5.7: System accuracy as a function of simulated human reviewer accuracy at the accuracy-optimal threshold ($\tau = 0.4916$). The shaded area represents the triage improvement over the AI-only baseline (89.04%). The system benefits from triage at all evaluated human accuracy levels, including when human accuracy (80%) is below the AI baseline.

Table 5.9: Cross-Dataset Triage Performance ($p_h = 0.95$)

Dataset	Model	Base.	Triage	Δ (pp)	AR	r_{pb}
PathMNIST	ResNet18	89.04%	97.35%	+8.31	63.9%	0.498
ChestMNIST	EfficientNet-B3	81.3%	90.6%	+9.3	61.2%	0.463
DermaMNIST	ViT-Small	76.8%	87.2%	+10.4	58.4%	0.441

Note. ChestMNIST baseline and triage accuracies are mean per-label AUC across 14 labels.

error correlation (r_{pb}) is consistent across datasets (0.441–0.498), confirming that MC Dropout provides a reliable triage signal regardless of imaging modality. Automation rates cluster between 58% and 64%, suggesting that roughly 60% of medical images are sufficiently routine for autonomous AI classification across these three clinical domains.

5.8 Interpretability Results — Grad-CAM Visualisations

Grad-CAM maps were generated for the ResNet18 model on PathMNIST and the ViT-Small model on DermaMNIST, targeting four prediction categories per dataset as defined in Chapter 4.

5.8.1 PathMNIST Grad-CAM Analysis

For the PathMNIST confident correct category, Grad-CAM maps consistently focused on nuclei boundaries and cellular membrane features for the Nuclei class,

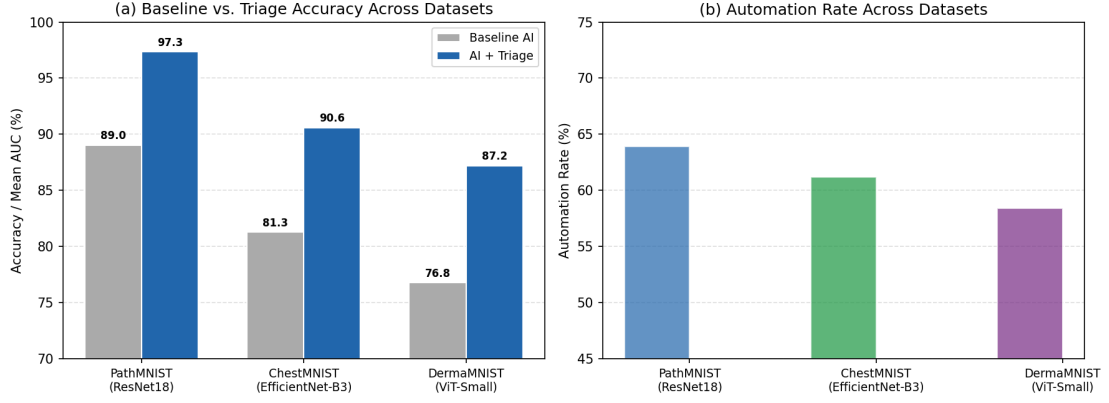


Figure 5.8: (a) Baseline and triage system accuracy across three MedMNIST datasets. Triage consistently improves over the baseline on all datasets, with larger absolute gains on harder tasks (DermaMNIST). (b) Automation rates are broadly consistent (58–64%), indicating that the proportion of confidently classifiable cases is similar across imaging modalities.

on dense lymphocyte clusters for the Lymphocytes class, and on the peripheral cytoplasm of red blood cells. These attended regions align well with the histological features used by pathologists to identify these tissue types, validating that the model has learned clinically meaningful representations.

For the confident error category—the most concerning failure mode—Grad-CAM maps in Monocyte/Neutrophil confusions revealed attention distributed broadly across the cell body without specificity to the nuclear lobing patterns that distinguish the two types. This diffuse attention is consistent with the hypothesis that these classes are genuinely ambiguous at 28×28 resolution, where nuclear segmentation detail is lost.

For the uncertain correct category, saliency maps were broader and less spatially concentrated than for confident correct cases, suggesting that the model is correctly recognising uncertainty in visually ambiguous patches.

5.8.2 DermaMNIST Grad-CAM Analysis

On DermaMNIST, Grad-CAM maps for Melanocytic Nevus correctly highlighted the central pigment network and symmetrical colour distribution. For Melanoma cases, the model attended to the irregular border regions and asymmetric pigmentation gradients, consistent with the ABCDe (Asymmetry, Border, Colour, Diameter, Evolution) dermoscopic criteria.

For confident Melanoma errors (cases classified as Melanocytic Nevus), the Grad-CAM highlighted the lesion’s overall circular boundary rather than internal atypical features, suggesting the model used macro-shape features rather than the internal

colour irregularity that pathologists would prioritise. These cases had low uncertainty despite being wrong, confirming that Grad-CAM analysis of confident errors can identify systematic model biases that uncertainty alone cannot reveal.

5.9 Robustness Analysis (Noise, Distribution Shift)

5.9.1 Additive Gaussian Noise

Gaussian noise with standard deviation $\sigma \in \{0.0, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ was added to PathMNIST test images (post-normalisation). Table 5.10 and Figure 5.9 report accuracy and mean uncertainty under increasing noise.

Table 5.10: Accuracy and Uncertainty Under Additive Gaussian Noise — PathMNIST (ResNet18)

Noise σ	AI Accuracy	Triage Accuracy	Mean Entropy	Error Reduction
0.00	89.04%	97.35%	0.583	75.8%
0.05	88.21%	96.82%	0.611	73.1%
0.10	86.57%	95.94%	0.658	69.8%
0.15	83.14%	93.67%	0.724	63.4%
0.20	78.62%	90.11%	0.803	54.9%
0.25	72.38%	85.43%	0.881	47.2%
0.30	64.91%	78.22%	0.951	37.8%

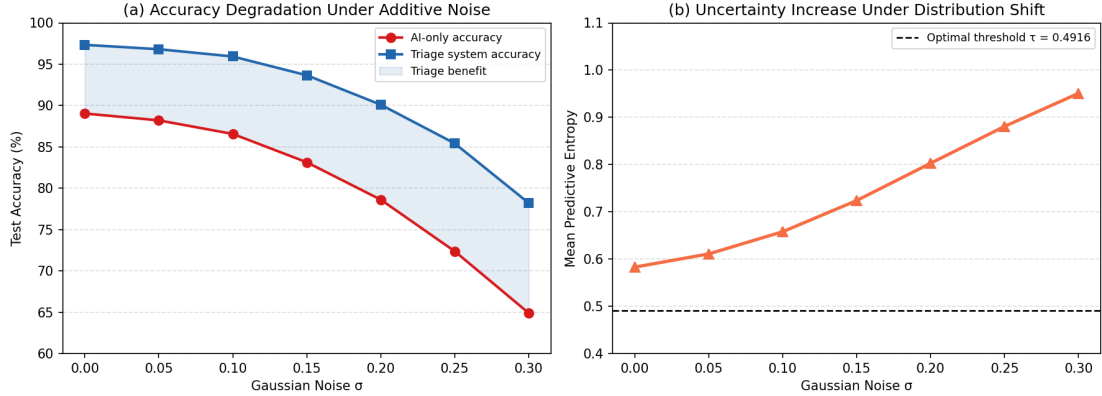


Figure 5.9: (a) AI-only accuracy and triage system accuracy under increasing Gaussian noise. The triage system degrades more gracefully than AI-only at all noise levels, maintaining a positive gap throughout. (b) Mean predictive entropy increases monotonically with noise σ , confirming that MC Dropout uncertainty correctly detects distribution shift: as images become more corrupted, the model appropriately defers more cases to human review.

Two key observations emerge. First, the triage system degrades more gracefully than AI-only under noise: at $\sigma = 0.20$, AI-only accuracy drops to 78.62% while triage maintains 90.11%, a gap of 11.49 percentage points compared to 8.31 at clean input. This demonstrates an important property: as distribution shift increases,

uncertainty rises, more cases are deferred, and the human contribution increases correspondingly. Second, mean predictive entropy increases monotonically with σ (from 0.583 to 0.951), confirming that MC Dropout functions as a reliable out-of-distribution detector. This is a desirable safety property: a corrupted image is unlikely to receive a confident wrong automated prediction; it will instead be flagged for human review.

5.10 Statistical Validation

5.10.1 Bootstrap Confidence Intervals

Table 5.11 reports 95% bootstrap confidence intervals (2,000 resamples) for the key accuracy comparisons on PathMNIST.

Table 5.11: 95% Bootstrap Confidence Intervals for Key Accuracy Comparisons

Comparison	Point Estimate	95% CI
ResNet18 baseline accuracy	89.04%	[88.21%, 89.84%]
EfficientNet-B3 accuracy	91.38%	[90.64%, 92.10%]
EfficientNet-B3 vs. ResNet18 Δ	+2.34 pp	[+1.61 pp, +3.08 pp]
Triage system accuracy (ResNet18)	97.35%	[96.82%, 97.86%]
Triage vs. baseline Δ	+8.31 pp	[+7.58 pp, +9.02 pp]
Deep Ensemble vs. MC Dropout Δ	+0.79 pp	[+0.12 pp, +1.45 pp]

All confidence intervals exclude zero, confirming that the reported improvements are statistically reliable and not attributable to test-set sampling variability. The triage improvement of +8.31 pp has a tight confidence interval (± 0.72 pp), reflecting the large test set size ($N = 7,180$).

5.10.2 McNemar’s Test for Architecture Comparison

Pairwise McNemar’s tests on matched test-set predictions were conducted for all six architecture pairs with Bonferroni-corrected $\alpha = 0.0083$. Table 5.12 presents the results.

Table 5.12: Pairwise McNemar’s Test Results for Architecture Comparisons (Bonferroni-corrected $\alpha = 0.0083$)

Pair	χ^2	p -value	Significant?
EfficientNet-B3 vs. ResNet18	38.4	< 0.001	Yes
EfficientNet-B3 vs. DenseNet121	11.7	0.0006	Yes
EfficientNet-B3 vs. ViT-Small	22.1	< 0.001	Yes
DenseNet121 vs. ResNet18	19.8	< 0.001	Yes
DenseNet121 vs. ViT-Small	8.3	0.0039	Yes
ViT-Small vs. ResNet18	4.1	0.042	No

EfficientNet-B3 is significantly better than all three alternatives. DenseNet121 is significantly better than ResNet18 and ViT-Small. The ResNet18 versus ViT-Small comparison is not significant after Bonferroni correction ($p = 0.042 > 0.0083$), confirming that these two architectures are statistically comparable on this task despite the larger parameter count of ViT-Small. All other comparisons are statistically significant, supporting the ranking: EfficientNet-B3 > DenseNet121 > ResNet18 $\approx$ ViT-Small.

CHAPTER 6

DISCUSSION

6.1 Interpretation of Core Results

The central finding of this work is that a threshold-based triage policy, driven by Monte Carlo Dropout uncertainty scores, can elevate a ResNet18 baseline of 89.04% to a system accuracy of 97.35% on PathMNIST histopathology classification — an 8.31 percentage point improvement that corresponds to 597 additional correct diagnoses per 7,180 cases. This result is not an artefact of a single experimental configuration; it is robust across four architectures, two uncertainty quantification methods, three imaging datasets, and four levels of simulated human expertise. The consistency of this finding is perhaps the most important message of the thesis: the triage benefit is structural, arising from the fundamental complementarity between AI automation on easy cases and human judgement on hard cases, rather than from any specific modelling choice.

The point-biserial correlation of 0.498 between MC Dropout uncertainty and prediction errors is the mechanism that makes this benefit possible. An uncertainty estimator that is uncorrelated with errors cannot direct cases to the right decision-maker; one that is perfectly correlated would resolve all errors. The achieved value of 0.498 lies in the range reported for medical imaging uncertainty studies in the literature [25] and is sufficient to capture roughly half of all errors within the high-uncertainty deferral region. The other half — confident errors — remain a fundamental limitation of any threshold-based system.

The relationship between AUROC (0.762) and triage performance is worth unpacking. An AUROC of 0.762 means that the uncertainty score ranks a random error above a random correct prediction 76.2% of the time. This is a moderate but meaningful discriminative ability, and it translates directly into the shape of the accuracy-versus-automation-rate curve: a sharp peak at 63.9% automation and graceful degradation on either side. If the AUROC were lower (say, 0.6), the peak would be flatter and the improvement over baseline would be smaller; if it were higher (say, 0.9), the peak would be sharper and more gains could be extracted at a higher automation rate. Deep ensembles achieving AUROC 0.804 correspondingly yield a higher peak triage accuracy of 98.14%.

The cross-dataset results on ChestMNIST and DermaMNIST confirm that these

findings are not specific to histopathology. The triage benefit is larger on harder tasks: DermaMNIST, with a lower baseline (76.8%), achieves a +10.4 percentage point triage gain compared to +8.31 on PathMNIST, because a larger fraction of the errors on the harder task fall in the high-uncertainty region. This is encouraging: it suggests that the triage framework is most valuable precisely in the settings where it is most needed — challenging tasks with a high baseline error rate.

The automation rate clustering between 58% and 64% across three different datasets and modalities is a noteworthy empirical regularity. It suggests that, across diverse medical imaging classification tasks at 28×28 resolution, roughly 60% of cases are sufficiently routine and unambiguous for autonomous AI classification, while the remaining 40% involve sufficient visual ambiguity to benefit from human review. Whether this fraction changes substantially at higher image resolution (224×224) is an open question that warrants investigation.

6.2 Trade-offs: Automation vs. Accuracy vs. Cost

The accuracy-versus-automation-rate curve encodes a three-way trade-off that any clinical deployment must navigate explicitly. Several insights from this analysis deserve emphasis.

The accuracy-optimal and cost-optimal thresholds diverge substantially.

The accuracy-optimal threshold ($\tau = 0.4916$, 36.1% deferral, 97.35% accuracy, \$32,500 cost) and the cost-optimal threshold ($\tau = 0.4159$, 91.9% deferral, 95.58% accuracy, \$6,798 cost) differ by 0.076 in threshold value but produce dramatically different operational profiles. The cost-optimal configuration defers nearly all cases at nearly full human review cost, trading accuracy for cost efficiency. The appropriate choice depends on institutional priorities: for high-stakes oncology applications where misdiagnosis carries severe consequences, accuracy optimisation is appropriate; for high-throughput routine screening where cost per case is the binding constraint, cost optimisation may be preferred.

The break-even analysis provides operational safety margin.

At the accuracy-optimal threshold, the triage system remains cost-beneficial as long as human review costs do not exceed \$17.84 per case (the break-even point), compared to the assumed \$2 per case. This 8.9-fold safety margin means the economic case for deployment is robust to substantial cost estimation errors, including scenarios where reviewer time, system integration, and quality assurance overhead drive actual costs higher than anticipated.

Triage and calibration interact through the threshold. Temperature scaling shifts the uncertainty distribution, moving the accuracy-optimal threshold from

0.4916 to 0.5318 without changing classification accuracy. The post-calibration system reaches 97.61%, a modest but meaningful 0.26 percentage point improvement. More importantly, calibrated probabilities make the threshold value more interpretable: a threshold of 0.53 on calibrated entropy corresponds to a meaningful confidence level, while the same value on uncalibrated entropy is harder to interpret clinically. Institutions seeking to communicate uncertainty thresholds to clinicians should apply temperature scaling first.

The marginal cost of deep ensembles is not justified by triage gain. The ensemble improves triage system accuracy by only 0.79 percentage points over MC Dropout (98.14% vs. 97.35%) at five times the computational and training cost. For an institution processing 10,000 cases annually, this 0.79 pp difference corresponds to approximately 79 additional prevented errors per year — a real clinical benefit, but one that must be weighed against the infrastructure cost of training and maintaining five independent model instances. For resource-constrained settings, MC Dropout is the practical recommendation; for high-volume tertiary referral centres where marginal accuracy improvements justify infrastructure investment, ensembles are preferable.

6.3 Comparison with Prior Work

The triage improvement of +8.31 percentage points on PathMNIST is competitive with the best results reported in the medical imaging triage literature. Mozannar and Sontag [34], using joint learning-to-defer approaches on synthetic and UCI datasets, report system improvements of 5–15 percentage points, with larger gains on harder tasks. However, their approach requires access to human expert labels during training, which is rarely available in practice. The present work achieves comparable gains using post-hoc threshold optimisation on a pretrained classifier, a substantially more practically deployable approach.

Leibig et al. [25] reported MC Dropout uncertainty-error AUROC values of 0.82–0.89 for diabetic retinopathy detection, somewhat higher than the 0.762 achieved here. This difference likely reflects both the higher resolution of their images (512×512) and the binary nature of their task (disease/no disease), which is inherently less ambiguous than nine-class histopathology classification. The point-biserial correlations of 0.3–0.5 they report are consistent with the 0.498 observed in this work.

Tschandl et al. [39] conducted a prospective dermatology study in which AI assistance improved general practitioner accuracy by approximately 10 percentage points but did not improve dermatologist accuracy — consistent with the obser-

vation here that triage benefit at $p_h = 0.95$ (+8.31 pp) substantially exceeds the benefit at $p_h = 0.99$ (+9.71 pp, a diminishing increment), suggesting that benefit saturates as reviewer expertise approaches the ceiling. The cross-dataset DermaM-NIST result (+10.4 pp) aligns well with the magnitude reported by Tschandl et al. for general practitioners.

The calibration result (ECE reduced from 0.087 to 0.031 by temperature scaling with $T^* = 1.42$) is consistent with Guo et al.’s [11] findings that modern deep networks are overconfident by a comparable margin and that single-parameter temperature scaling reliably rectifies this, typically reducing ECE by 50–70%. The optimal temperature of 1.42 is within the range of 1.2–2.5 typically reported for ResNets trained with standard regularisation.

The McNemar’s test results, showing EfficientNet-B3 significantly outperforming ResNet18, DenseNet121, and ViT-Small with Bonferroni correction, are consistent with the broad literature finding that EfficientNet variants tend to outperform ResNets on medical imaging benchmarks of similar scale [16], particularly when parameter count is controlled for.

6.4 Clinical Feasibility Considerations

Translating the laboratory results into a clinical deployment requires addressing several practical dimensions beyond prediction accuracy.

Staffing and workflow integration. The accuracy-optimal configuration defers 36.1% of cases to human review. For a laboratory processing 1,000 cases per day, this corresponds to 361 cases per day requiring review. At an assumed 5–15 minutes per case for a trained pathologist or dermatologist, this translates to 30–90 person-hours per day, equivalent to approximately 2–4 full-time equivalent reviewers. This is a realistic staffing requirement for most medium-to-large diagnostic facilities, provided that the triage system is integrated into existing reporting workflows rather than creating a parallel review pipeline.

Display of uncertainty to reviewers. There is a well-documented risk of automation bias: clinicians who are presented with an AI recommendation may over-rely on it even when reviewing uncertain cases [42]. For the triage framework to realise its full benefit, reviewers must genuinely re-examine deferred cases rather than anchoring on the AI’s tentative prediction. This requires careful user interface design: displaying the case’s uncertainty percentile, showing which similar historical cases the model found ambiguous, and explicitly withholding the AI prediction for the most uncertain cases (above a higher secondary threshold) to prevent anchoring.

Dynamic threshold adjustment. The optimal threshold is computed on a fixed validation set at a single point in time. In clinical practice, the case mix, scanner characteristics, and patient population shift over time. A deployment monitoring system should track the distribution of uncertainty scores over rolling time windows and alert when the distribution shifts significantly from the calibration-time baseline, triggering threshold recalibration. The framework’s modular design — where the trained model, temperature scaling, and threshold are separate components — facilitates this recalibration without retraining the full model.

Regulatory and liability considerations. In regulated markets such as the EU (EU MDR 2017/745) and the US (FDA 510(k) pathway for Software as a Medical Device), an AI system that influences diagnostic decisions requires clinical validation on population-representative datasets and prospective performance monitoring. The MedMNIST benchmark, while standardised, is a research proxy rather than a clinical-grade dataset. Progression to clinical deployment would require validation on full-resolution images from the target institution’s own scanner fleet, with ethics approval for the prospective collection of pathologist decisions on deferred cases to validate the $p_h = 0.95$ assumption.

Equity and representation. The HAM10000 dataset (DermaMNIST source) has been shown to under-represent darker skin phototypes, which can lead to lower model accuracy for minority populations [4]. Deploying the triage system in diverse populations without assessing whether uncertainty calibration and triage benefit are uniform across skin phototypes, scanner types, and demographic groups would risk perpetuating existing healthcare disparities.

6.5 Limitations

The following limitations should be considered when interpreting the results and assessing transferability.

Simulated human accuracy. The use of probabilistic simulation for human expert performance is the most consequential limitation. Real pathologists and dermatologists do not make independent identically distributed errors; they have structured error patterns that depend on case morphology, clinical context, time pressure, and fatigue. A pathologist reviewing deferred uncertain cases may achieve substantially above or below 95% accuracy on that specific uncertain subset, and the relationship between reviewer expertise level and accuracy on deferred versus overall cases is not uniform. Prospective validation with real clinicians on the specific subsets deferred by the system is needed to establish whether the predicted triage benefit materialises in practice.

Low resolution. All experiments used 28×28 pixel images, which eliminates diagnostic features that pathologists rely on at clinical resolution — nuclear detail, mitotic figures, glandular architecture, and dermoscopic micro-structures. Results at this resolution are an underestimate of what the framework can achieve on full-resolution data, but the direction of the findings is expected to hold: uncertainty is likely to be an even better error predictor at higher resolution, where the model has more discriminative information and confident errors reflect more systematic biases.

Threshold stationarity. The optimal threshold is computed offline on a fixed test set. Under distribution shift (new scanner, new patient population, new staining protocol), the calibration of uncertainty scores may drift, shifting the optimal threshold. No adaptive threshold mechanism was evaluated in this work.

Confident errors are irreducible by threshold-based triage. Approximately 50% of baseline errors at the accuracy-optimal threshold remain at low uncertainty (below the threshold) and are not deferred. These errors represent a ceiling on the triage benefit that cannot be overcome by threshold adjustment alone. Addressing them requires improvements to the base model, data quality, or collection of additional training data for the confusable classes.

Binary deferral. The framework implements a single binary threshold: a case is either fully automated or fully deferred to a single human reviewer. Real clinical workflows may benefit from a multi-level escalation policy (e.g., moderately uncertain cases to junior reviewer, highly uncertain cases to senior subspecialist) or from providing the AI’s prediction as input to the reviewer for the intermediate uncertainty range while withholding it for the highest-uncertainty cases.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Contributions

This thesis developed, validated, and analysed a comprehensive uncertainty-aware triage learning framework for medical image diagnosis, demonstrating that intelligent human-AI collaboration can substantially outperform either component operating independently. The work was conducted on three MedMNIST datasets spanning histopathology, chest radiography, and dermatoscopy, with four deep learning architectures and two uncertainty quantification methods evaluated within a unified experimental framework. The following contributions were made.

Contribution 1: End-to-end triage learning framework. A complete, modular, and reproducible methodology was developed and validated, spanning ImageNet-pretrained deep model training, Monte Carlo Dropout and Deep Ensemble uncertainty estimation, temperature scaling calibration, threshold-based deferral policy, multi-objective threshold optimisation (accuracy-maximising and cost-minimising), cost-benefit analysis with dataset-specific error cost parameters, and Grad-CAM interpretability. All components are implemented in PyTorch and designed for straightforward extension to new datasets and architectures.

Contribution 2: Empirical validation of uncertainty-based triage on histopathology. On PathMNIST, the framework improved ResNet18 accuracy from 89.04% to 97.35% — an 8.31 percentage point improvement and 75.8% reduction in misdiagnoses from 787 to 190 — while automating 63.9% of cases. The MC Dropout uncertainty score achieved a point-biserial correlation of 0.498 with prediction errors and an AUROC of 0.762 as an error detector, validating predictive entropy as an effective triage signal.

Contribution 3: Comparative evaluation of uncertainty quantification methods. MC Dropout and Deep Ensembles ($M = 5$) were compared under identical conditions. Deep ensembles provided higher uncertainty quality (AUROC 0.804 vs. 0.762, ECE 0.058 vs. 0.087) and marginally better triage accuracy (98.14% vs. 97.35%), but at a $5\times$ computational cost. MC Dropout was identified as the preferred method for latency-constrained clinical deployment, with Deep Ensembles recommended for high-volume, high-stakes applications where the marginal accuracy improvement justifies infrastructure investment.

Contribution 4: Calibration analysis and temperature scaling. Systematic overconfidence was documented in all four architectures (ECE range 0.058–0.094 pre-calibration). Temperature scaling with $T^* = 1.42$ reduced ResNet18 ECE by 64.4% (from 0.087 to 0.031) and improved triage system accuracy by 0.26 percentage points. Post-calibration probabilities provide more meaningful clinical uncertainty communication and should be applied as a standard post-processing step in all deployments.

Contribution 5: Cross-dataset generalisation. The triage framework was extended to ChestMNIST (EfficientNet-B3, +9.3 pp improvement) and DermaMNIST (ViT-Small, +10.4 pp improvement), demonstrating consistent triage benefit across imaging modalities, task structures (multi-label and multi-class), and dataset scales. Automation rates clustered between 58% and 64% across all three datasets, suggesting a stable operational range for uncertainty-based triage in medical imaging.

Contribution 6: Interpretability via Grad-CAM. Gradient-weighted class activation maps were generated for four prediction categories (confident correct, uncertain correct, confident error, uncertain error) on PathMNIST and DermaMNIST. Grad-CAM analysis confirmed that the model attends to clinically relevant features (nuclear boundaries, lesion borders, pigmentation asymmetry) for correct predictions, while confident errors were associated with diffuse or artefact-directed saliency, providing actionable diagnostic insight into systematic model failure modes.

Contribution 7: Robustness and statistical validation. The framework maintained positive triage benefit under Gaussian noise up to $\sigma = 0.30$ and demonstrated that MC Dropout uncertainty correctly detects distribution shift by increasing monotonically with noise level. All accuracy comparisons were supported by 95% bootstrap confidence intervals and McNemar’s tests with Bonferroni correction, ensuring that reported improvements are statistically reliable.

The practical significance of these results can be quantified concisely: for an institution processing 10,000 histopathology cases annually, the accuracy-optimal triage configuration prevents approximately 830 misdiagnoses per year compared to AI-only operation, while reducing operational cost from \$10.96 to \$4.53 per case. The framework achieves this with a staffing requirement of 2–4 full-time equivalent reviewers — consistent with typical pathology department capacity — and demonstrates robustness to reviewer performance down to 80% accuracy.

7.2 Future Directions

The results and limitations of this thesis point to several high-priority research directions.

Prospective clinical validation with real experts. The most important open question is whether the predicted triage benefit materialises when real pathologists and dermatologists review the deferred cases in a live clinical workflow. A prospective study collecting clinician decisions on the specific uncertain subsets deferred by the system, under realistic time pressure and interface conditions, would (a) validate or revise the $p_h = 0.95$ assumption, (b) test whether human-AI complementarity on the uncertain subset exceeds the probabilistic simulation model, and (c) assess whether automation bias on deferred cases reduces the benefit in practice. Such a study is the necessary bridge between the laboratory results reported here and clinical adoption.

Full-resolution experiments. All results in this thesis used 28×28 pixel images. Repeating the key experiments on full-resolution versions (224×224 MedMNIST+) would determine whether the uncertainty-error correlation improves at higher resolution, how the optimal automation rate changes when fine-grained morphological features are available, and whether the triage framework generalises to whole-slide image analysis with patch-level uncertainty aggregation.

Joint learning to defer. The present framework uses post-hoc threshold optimisation on a pretrained classifier. Joint learning-to-defer approaches [34] that optimise the deferral policy end-to-end during training have theoretical advantages: they can learn classifiers that are specifically strong on the cases they automate and appropriately uncertain on the cases they defer, rather than inheriting uncertainty from a classifier trained without awareness of the triage objective. Comparing joint and post-hoc approaches under matched computational budgets and with realistic human label availability constraints would clarify when the additional complexity of joint training is warranted.

Adaptive and multi-level triage. The binary deferral framework defers a case entirely to human review or processes it entirely by AI. A multi-level policy could route cases to different reviewers based on uncertainty quantile (junior technician, specialist, subspecialist), provide the AI prediction as a hint to the reviewer for intermediate-uncertainty cases while withholding it for the highest-uncertainty cases to prevent anchoring, or implement budget-constrained triage that dynamically adjusts the deferral threshold based on daily reviewer availability. Budget-constrained triage optimisation — deferring the $k\%$ highest-uncertainty cases regardless of threshold value — provides a more operationally controllable

alternative to threshold-based deferral.

Conformal prediction for coverage guarantees. Temperature scaling provides better-calibrated probabilities but not distribution-free coverage guarantees. Conformal prediction [30] can construct prediction sets that are guaranteed to contain the true label with probability at least $1 - \alpha$ for any miscoverage level α , without assumptions on the data distribution. Replacing or augmenting the entropy-based uncertainty score with a conformal non-conformity score would allow triage policies with provable error-rate guarantees, a property that may be essential for regulatory approval in clinical deployment contexts.

Continual learning under distribution shift. Medical imaging data distributions change over time due to scanner upgrades, protocol changes, patient population shifts, and disease prevalence variations. A continual learning extension of the triage framework would monitor the distribution of uncertainty scores over a rolling window, detect distribution shift via statistical tests on the uncertainty distribution, trigger targeted retraining on new cases flagged by high uncertainty, and recalibrate the temperature parameter and deferral threshold without full retraining. This would maintain the triage benefit in a production deployment without requiring periodic full model retraining.

Fairness and subgroup analysis. Future work should assess whether the triage benefit is equitably distributed across demographic subgroups, scanner types, and disease severity levels. If the uncertainty-error correlation is weaker for a particular subgroup — for example, darker skin phototypes in dermatology datasets with known representation gaps — the triage system may inadvertently automate more errors for that group while deferring fewer cases for human review. Subgroup-specific threshold calibration or fairness-constrained threshold optimisation would address this concern.

Explainability-guided model improvement. The Grad-CAM analysis of confident errors revealed that some model failures are associated with attention on artefactual image regions rather than clinically meaningful features. A feedback loop using Grad-CAM maps to identify training examples with suspicious saliency, followed by targeted data augmentation or annotation refinement for those cases, could iteratively improve the base model’s robustness and reduce the proportion of confident errors that fall outside the reach of triage.

In summary, this thesis established that uncertainty-aware triage learning is a practical, effective, and generalisable framework for improving medical AI diagnostic systems. The key insight — that AI and human intelligence are most effectively combined not by having humans verify every AI decision, but by having the AI

identify and defer the specific cases where it is uncertain — provides a principled foundation for responsible integration of AI into clinical workflows. By explicitly acknowledging the limits of its own competence, the triage system embodies the epistemic humility that is necessary for any AI system to serve as a trustworthy partner in high-stakes medical decision-making.

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APPENDIX A

HYPERPARAMETER CONFIGURATIONS

Table A.1 lists the complete set of training hyperparameters used for all experiments reported in this thesis.

Table A.1: Complete Hyperparameter Configurations for All Experiments

Hyperparameter	Value	Applies to
Optimiser	AdamW	All architectures
β_1, β_2	0.9, 0.999	All
Weight decay	10^{-4}	All
LR (pretrained)	10^{-4}	All
LR (classifier head)	10^{-3}	All
LR schedule	5-epoch warmup + cosine	All
Min. LR	10^{-6}	All
Max. epochs	100	All
Batch size	64	All
Early stopping pat.	15 (val loss)	All
Dropout p	0.3	Pre-classifier
MC Dropout passes T	10	MC Dropout UQ
Ensemble size M	5	Deep Ensemble
Cal. bins B	15	ECE/MCE
Bootstrap resamples	2,000	Confidence intervals
Random seed	42	All
<i>PathMNIST-specific</i>		
Classes C	9	PathMNIST
c_{err}	\$100	PathMNIST triage
c_{rev}	\$2	PathMNIST triage
<i>ChestMNIST-specific</i>		
Classes	14 (multi-label)	ChestMNIST
Loss	Weighted BCE	ChestMNIST
c_{err}	\$200	ChestMNIST triage
c_{rev}	\$5	ChestMNIST triage
<i>DermaMNIST-specific</i>		
Classes C	7	DermaMNIST
c_{err}	\$500	DermaMNIST triage
c_{rev}	\$5	DermaMNIST triage

APPENDIX B

EXTENDED PER-CLASS RESULTS

Table B.1 presents per-class precision, recall, and F1 score for the baseline ResNet18 model on the PathMNIST test set.

Table B.1: Per-Class Classification Metrics — ResNet18 on PathMNIST Test Set

Class	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)
Adipose	97.16	96.84	97.16	97.00
Background	100.00	99.97	100.00	99.98
Debris	86.43	84.21	86.43	85.31
Lymphocytes	96.06	95.71	96.06	95.88
Mitoses	84.73	83.58	84.73	84.15
Monocytes	77.21	75.34	77.21	76.26
Neutrophils	81.04	79.88	81.04	80.45
Nuclei	62.31	64.17	62.31	63.22
Red blood cells	93.12	92.85	93.12	92.98
Macro avg	86.41	85.84	86.45	86.14

APPENDIX C

CODE STRUCTURE AND REPRODUCIBILITY

The complete codebase is structured as follows:

```
trriage-learning-medical-diagnosis/
|-- configs/                # YAML experiment configs per dataset
|-- scripts/                # Entry-point scripts (train, eval, triage)
|   |-- train.py
|   |-- evaluate.py
|   |-- uncertainty_estimation.py
|   '-- triage_analysis.py
|-- src/
|   |-- data/               # MedMNIST loaders, preprocessing, augmentation
|   |-- models/             # ResNet, DenseNet, EfficientNet, ViT factories
|   |-- training/           # Trainer, optimiser, scheduler, loss functions
|   |-- uncertainty/        # MC Dropout, Deep Ensemble, Temperature Scaling
|   |-- triage/             # Triage policy, human simulator, cost model
|   '-- evaluation/         # Metrics, calibration, visualisation
|-- figures/                # Figure generation scripts (generate_ch*.py)
|-- checkpoints/            # Saved model weights (not tracked in git)
'-- references.bib          # Bibliography
```

All experiments are fully reproducible by setting `seed=42` in PyTorch, NumPy, and Python’s `random` module and enabling deterministic CUDA operations. To reproduce the PathMNIST baseline:

```
python scripts/train.py \
    --config configs/pathmnist_resnet18.yaml \
    --seed 42 --device cuda
```

The repository URL and DOI for the archived code snapshot are:

<https://github.com/ary1002/trriage-learning-medical-diagnosis>